

EnergyAustralia's FY11 Network Pricing Proposal

April 2010 (revised 18 June)



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¹ Attachment C corrected on 25 June 2010

ENERGYAUSTRALIA'S FY11 NETWORK PRICING PROPOSAL

PART 1 - APPROACH TO PRICE SETTING

1. Introduction and Overview

Introduction

EnergyAustralia's transmission and distribution services are subject to price regulation. The AER makes a five yearly determination, which sets a limit on the increase in distribution and transmission prices (X-factors) that can be applied by EnergyAustralia. The price limit is based on providing EnergyAustralia with a revenue requirement that is sufficient to recover efficient capital and operating costs.

On 30 April 2009 the Australian Energy Regulator (AER) made a determination on the X-factors to apply to EnergyAustralia's distribution and transmission services for the 2009-14 regulatory period.

Subsequent to this decision, EnergyAustralia (and the other two NSW distribution network businesses - Integral Energy and Country Energy) sought review of elements of the AER's determination. On 25 November 2009, the Australian Competition Tribunal varied the AER Determination by increasing the nominal vanilla WACC (that is, the rate of return on investments) from 8.78% to 10.02%. The Australian Competition Tribunal also increased EnergyAustralia's operating expenditure allowance by \$4.5 million.

The Tribunal's determination varied the X-factors to apply to EnergyAustralia's distribution and transmission services for the last four years of the regulatory period, from that which was published by the AER in its April 2009 determination. The amended X-factors are set out below:

Table 1: EnergyAustralia's Annual X Factor – Remainder of Regulatory Period

	FY10	FY11	FY12	FY13	FY14
Forecast CPI	2.47%	2.47%	2.47%	2.47%	2.47%
Distribution	-17.86%	-18.18%	-18.18%	-18.18%	0.77
Transmission	-7.77%	-18.46%	-18.46%	-18.46%	-2.02

In addition to its decision on the X-factors to apply to distribution and transmission services, the AER must also make a decision on the price control to apply to public lighting services. EnergyAustralia sought review of the AER's decision of April 2009. As a result of this review, the Australian Competition Tribunal remitted the decision back to the AER for re-determination. The AER's re-determination for EnergyAustralia's public lighting services was published on 15 April 2010, and is discussed in Chapter 7 of Part 1 and the customer bill information and proposed prices for public lighting services are provided in Attachment B and C, respectively.

As a result of the new overall X factor increases for distribution prices and transmission charges in FY11 and the increase in the contribution to the NSW State Government's Climate Change Fund, the total impact on the overall network component of the annual electricity bill for customers on the inclining block tariffs (IBT) in FY11 is summarised below:

- a typical domestic customer² on IBT is estimated to receive a nominal \$1.58 per week increase in FY11.

- a typical small business customer³ on IBT is estimated to receive a nominal \$3.54 per week increase in FY11.

Document structure

This document is EnergyAustralia's Pricing Proposal for FY11. It sets out how we proposed to set network tariffs in FY11 to recover the efficient cost of owning, operating, maintaining and investing in the distribution and transmission networks.

This document also provides external stakeholders with the proposed prices for EnergyAustralia's alternative control (public lighting) services for FY11.

This proposal is submitted for review and approval by the AER as required by clause 6.8.2(a)(1) of the transitional chapter 6 of the National Electricity Rules (the Rules).

The proposal is split in two parts:

- Part 1 outlines EnergyAustralia's method and strategy in setting network prices. It is written with consumers and other interested parties in mind, to assist in understanding how network tariffs are calculated.

² For an average domestic customer on the inclining block tariff who consumes 5,200 kWh per annum, and does not have controlled load hot water. Increase excludes GST. Retail related cost increases not included.

³ For an average small business customer on the inclining block tariff who consumes 10,500 kWh per annum. Increase excludes GST. Retail related cost increases not included.

Part 2 is more technical in nature as it has written specifically to demonstrate that EnergyAustralia's proposed network prices for FY11 comply with the obligations of the Rules. It also has a discussion on EnergyAustralia's public lighting prices for FY11. Part 2 is structured to allow ready assessment of compliance by the AER.

While every attempt has been made to write Part 1 of this document in a style easily understood by external stakeholders that are not familiar with network pricing and regulatory concepts, it has been necessary to introduce the reader to some technical terms. To assist in understanding these technical terms, a comprehensive glossary section has been added to the end of Part 1.

2. Background

This section aims to provide an understanding of the historical context of EnergyAustralia's network prices for FY11 by discussing the progress made in reforming network tariffs over the past five years, the recent regulatory decisions by the Australian Energy Regulator (AER) for distribution and transmission network revenue and prices, and variations to this decision ordered by the Australian Competition Tribunal.

EnergyAustralia's distribution and transmission services are subject to price regulation by the Australian Energy Regulator under the Rules known as the 'Transitional Chapter 6 Rules'. The Rules apply for the 2009-14 regulatory period and are set out in Appendix 1 of Chapter 11 of the National Electricity Rules made under the National Electricity Law.

On a five yearly basis, the AER makes a determination which sets a limit on the price increases that we can charge customers (referred to as an X-factor for each year). The X-factors enable EnergyAustralia to recover an annual revenue requirement. The revenue requirement is based on the AER's appraisal of the efficient costs faced by EnergyAustralia for the five year period. As discussed below, distribution and transmission services are subject to different types of price control, and each have a set of X-factors.

Weighted Average Price Cap (WAPC)

EnergyAustralia's distribution prices are governed by a Weighted Average Price Cap (WAPC). The WAPC is a price cap where a limit is applied to the annual movement in distribution prices. The limit applied under either a revenue or price cap is referred to as the X factor and could be positive (price or revenue reduction) or negative (price or revenue increase), refer to Section 3 of Part 1 of this document for a more detailed explanation of the WAPC constraint.

Under the WAPC, distributors need to ensure that their distribution charges are as cost reflective as possible. Otherwise, distribution prices may not generate sufficient revenue to cover the efficient cost of distribution service provision in the event of unanticipated growth in peak demand for network capacity, which drives the need for additional investment in network augmentation.

By improving the efficiency of network tariffs, customers are likely to better understand the economic cost of their network use. To the extent that customers do not value their network use more than the economic cost, customers will respond by changing their network use to eliminate this economic waste. According to economic theory, this type of customer response promotes the optimal allocation of society's scarce resources. One consequence of the WAPC is that EnergyAustralia is

exposed to distribution revenue risk arising from unanticipated variations in electricity consumption, which is discussed in more detail below.

Revenue Cap

EnergyAustralia's transmission service prices are governed by a revenue cap. A revenue cap limits the amount of revenue that can be earned each regulatory year. This form of regulation provides EnergyAustralia with certainty that actual transmission costs incurred will be recovered from network customers, as any under (over) recovery of these transmission costs in a given year is allowed to be recovered in the following year (plus interest accrued) by adjusting transmission use of system charges in the subsequent period.

Electricity consumption trends

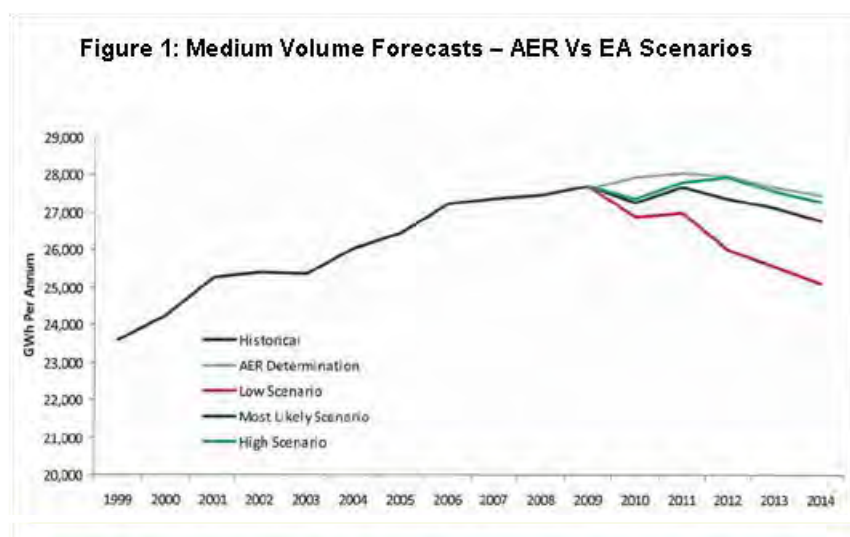
Every five years, the AER needs to decide the distributor's forecast electricity volumes for the subsequent 5-year regulatory period. These 5-year forecasts are used to convert the allowable distribution revenue into a 5-year distribution price path, which in turn is used to calculate the cap on distribution price movements at each 1 July price reset (refer to X factors (see Table 1 above for EnergyAustralia's X factors and Table 2 for TransGrid's X factors). Hence the AER plays a critical role in determining the level of risk faced by EnergyAustralia in earning its allowable distribution revenue over the 5-year regulatory period in question.

Importantly, EnergyAustralia is not only exposed to forecast volume risk at time of the setting of X factors every five years. Volume risk also arises from annual variation in electricity consumption volume due to such factors as unexpected weather, the take up of appliances such as air conditioners, or factors such as an economic downturn.

This volume risk translates into a distribution revenue risk because network prices are set on an annual basis, using the predetermined X factors, based on the most recent complete financial year of consumption. For example, the WAPC governing the setting of distribution tariffs in FY11 is based on

the audited volumes for FY09. In simple terms, this means that if actual consumption in FY11 is lower than the audited consumption in FY09, it is likely that distribution revenue in FY11 will fall short of expectations.

A summary of the differences between the overall forecast volumes used to set the X factor in the AER determination and a feasible range of forecasts based on more recent data is show in the following figure.



The future uncertainty in EnergyAustralia's volume environment reflected in the above figure highlights the need for EnergyAustralia to continue to improve the cost reflectivity of network tariffs. It also highlights the importance of continued research into the underlying drivers of the electricity consumption behaviour of households and business in EnergyAustralia's network area.

AER Final Decision (pre-Tribunal)

As discussed above, EnergyAustralia's distribution revenue and price paths are regulated by the AER. These are reviewed every five years. The most recent regulatory review concluded in April 2009, when the AER released its final determination for distribution and transmission. EnergyAustralia sought review from the Australian Competition Tribunal on elements of the AER's 2009 final determination, as set out below.

Tribunal Outcome

On 25 November 2009 the Australian Competition Tribunal made orders varying the AER's distribution determination for 2009–14 which included:

- determining the weighted average cost of capital (WACC) to be based on an August–September 2009 averaging period for the 10-year bond rates. This results in increasing the nominal vanilla WACC for EnergyAustralia to 10.02 per cent (from 8.78 per cent).
- increasing the controllable operating expenditure allowance for EnergyAustralia by \$4.5 million to \$2582 million (\$2008–09)

The Tribunal's determination varied the X-factors to apply to EnergyAustralia's distribution and transmission services for the last four years of the regulatory period. The X factors for EnergyAustralia's distribution and transmission services during the 2009–14 regulatory period are set out below, in addition to the respective annual revenue requirements.

Table 2: Australian Competition Tribunal Outcome for EnergyAustralia

EnergyAustralia	FY10	FY11	FY12	FY13	FY14
Distribution X factors	-17.86%	-18.18%	-18.18%	-18.18%	0.77%
Transmission X factors	-7.77%	-18.46%	-18.46%	-18.46%	-2.02%
Annual revenue requirement (distribution) \$m	1330.5	1519.6	1702.8	1903.1	2059.3
Annual revenue requirement (transmission) \$m	146.7	182.8	206.6	238.4	237.6

Public Lighting Services

EnergyAustralia's public lighting services regulated by the AER through a price control mechanism that is separate to that for distribution services, however it is reviewed every five years along side the review of the distribution price control. The most recent regulatory review concluded in April 2009, when the AER released it's final determination for distribution and transmission.

On 25 November 2009 the Australian Competition Tribunal made orders varying the AER's distribution determination for public lighting charges over 2009-10 to 2013-14, which included:

- determining the appropriate cost of capital to be consistent with the that for distribution services.
- remitting the public lighting decision back to the AER for it to remake its decision in relation to the efficient public lighting operating expenditure and correct any errors it identified.
- requiring the AER to review its decision on EnergyAustralia's public lighting asset value.

On 15 April 2010, the AER completed its review of public lighting prices and revenues. The AER's determination is on its website and chapter 9 provides an overview of the prices and charges that will result from that determination.

TransGrid

EnergyAustralia's network prices not only recover the efficient cost of operating, maintaining and augmenting EnergyAustralia's distribution and transmission assets, but they also include recovery for a portion TransGrid's transmission costs.

TransGrid is the owner, operator and manager of the NSW high voltage network, connecting generators, distributors and major end users in New South Wales. TransGrid's network comprises of almost 12,500 kilometres of high voltage transmission line and underground cables up to a voltage of 500kV. It also has six interconnectors to Victoria and Queensland's transmission network. While 60 percent of NSW energy is generated west of the Great Dividing Range it must be delivered to the east coast, where most of the state's load is located. The Sydney, Wollongong and Newcastle areas consume 75 percent of the energy.

TransGrid also received a five-year revenue determination from the AER in April 2009. TransGrid operates under a revenue cap, which provides more revenue certainty than the price cap governing EnergyAustralia's distribution prices. TransGrid also sought review of elements of the AER's determination, and as a consequence the Australian Competition Tribunal decided to vary the maximum allowed revenue (MAR) that was set out in the AER's April 2009 determination for TransGrid. The smoothed MAR for TransGrid is set out below.

Table 3: Australian Competition Tribunal Outcome for TransGrid

TransGrid	FY10	FY11	FY12	FY13	FY14
Max. Allowed Revenue (\$m)	678.4	734.2	794.5	859.8	930.5

Some of the reasons for the recent significant increases in TransGrid's revenue recovery are as follows:

- increases in the regulated revenue caps for NSW transmission businesses for the next pricing period;
- a continuing fall in the value of National Electricity Market trading residues returned to customers as a reduction in transmission charges; and
- any pass through adjustments that have been approved by the AER.

TransGrid has significantly rebalanced transmission charges for FY11 by removing the peak and shoulder energy prices and more than doubling the demand

charge. Given the significance of this rebalancing of TransGrid's transmission charges, EnergyAustralia has decided to transition it's transmission tariffs to cost reflective over a reasonable period where required to avoid imposing unacceptable price shocks on individual customers in FY11.

General Overview

The revenue stream for the EnergyAustralia network business is derived from our customer base in the proportions shown in figures 2a and 2b.

Figure 2a: FY11 Revenue by Customer Segment

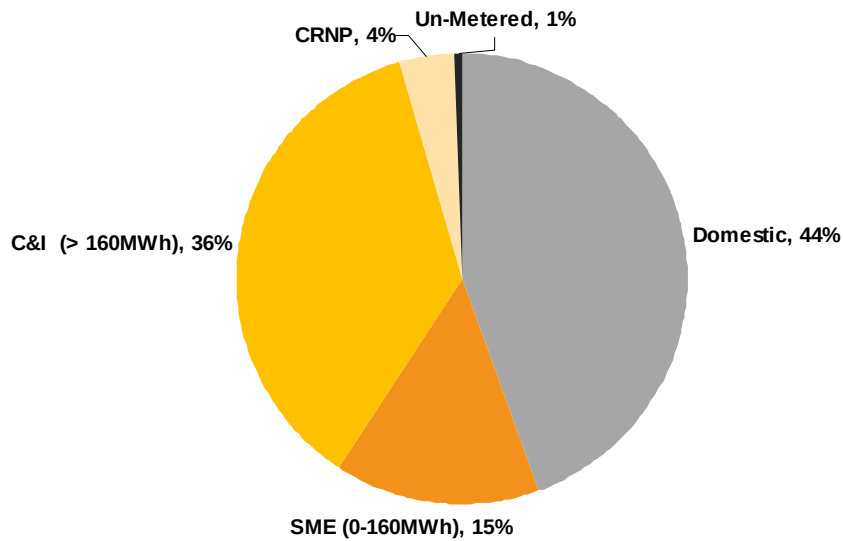
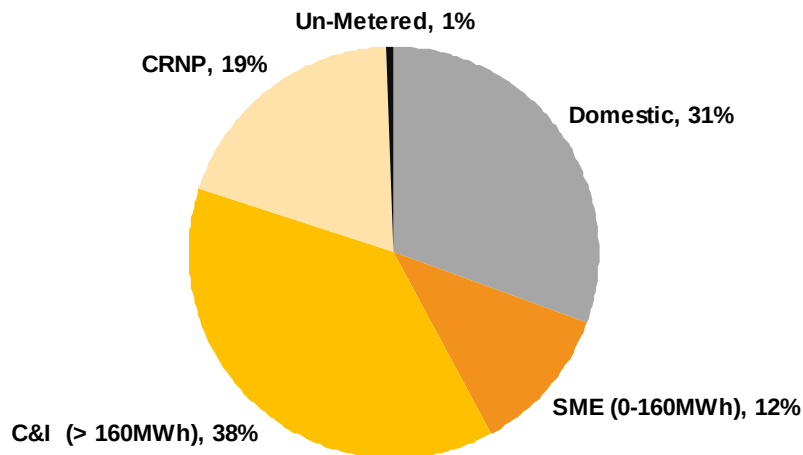


Figure 2b: FY11 Energy Consumption by Customer Segment

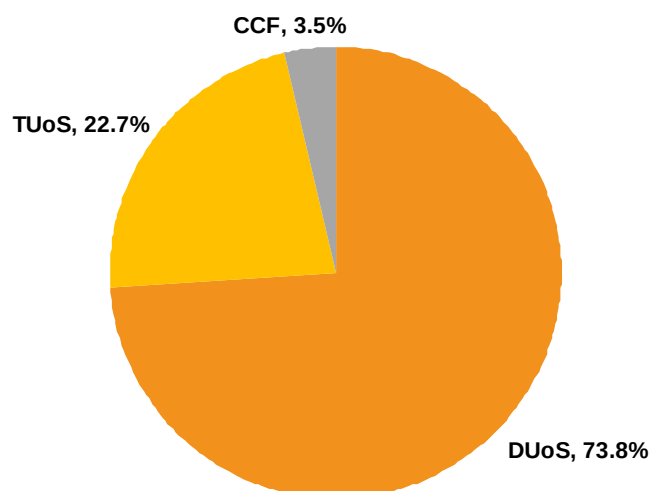


As can be seen from the above figures, EnergyAustralia's largest customers on a Cost Reflective Network Price (CRNP) tariff represent 19% of energy volumes but only represent 4% of revenues. Conversely, domestic customers only consume about a third of the total energy volumes, but account for 44% of revenue. This is reflective of the amount of assets required to service these customers. For instance, large customers, though

they use large volumes, are connected high up in the distribution network, whereas smaller customers use the electricity network all the way down to their LV connection. Therefore the cost reflective price is therefore higher for low voltage connected customers, compared to the cost reflective price for customers connected at either the high voltage or sub-transmission voltage level.

Finally, EnergyAustralia's network revenue is comprised of a number of components, as summarised in the following figure:

Figure 3: Composition of FY11 Revenue



As shown above, the cost of owning, operating, maintaining and investing in the distribution network is the largest component of annual network revenue accounting for around 74%. Included in the distribution revenue are earnings derived from a range of ancillary activities, such as extra meter readings, new connections, answering conveyancing enquires etc. This accounts for approximately 1% of distribution revenue.

The second largest component is the recovery of the cost of transmission network services⁴, which accounts for around 23% of annual network revenue. EnergyAustralia is also required to make a contribution to the NSW Government Climate Change Fund, which is used to promote energy efficiency and renewable energy technology. This contribution accounts for only 3% of annual network revenue.

⁴ The transmission network connects large generators together, and transports the energy they produce to distributors.

3. Pricing Methodology

The following discussion outlines how EnergyAustralia sets network charges, by providing an understanding of the four basic steps in setting of network tariffs, as summarised below:

- Identify a range of objectives for network tariffs to achieve.
- Develop pricing strategies that aim to achieve these objectives in a manner that strikes an appropriate balance between financial, economic and equity considerations.
- Establish an equitable amount of revenue to recover from each tariff class based on the users pay principle.

- Set network tariffs in accordance with the above steps and in compliance with price limits and the weighted average price cap imposed by AER.

Pricing Objectives

The network prices are set each year to achieve a range of economic and non-economic objectives, as summarised in the following table.

Table 4: EnergyAustralia's Network Pricing Objectives

Objective		Description/Principle
Achieve economic efficiency by providing customers with prices reflective of economic cost		Efficient network tariffs signal to retailers the economic cost of providing network capacity to end-customers at different times.
Achieve regulatory outcomes		Network tariffs will be set in accordance with the National Electricity Rules and the AER Determination. This means that distribution prices should, depending on volume considerations, generate revenues that cover the efficient cost of owning, maintaining, operating and augmenting the network.
Satisfy customers	Comprehensible	Network tariffs should be designed to: • Be as simple as possible so that customers are able to understand the bill implications of usage decisions. • Result in customers connected at the same level of the network with similar usage patterns receiving similar network charges. • Provide certainty of price movements, which in practice means that in some situations distribution prices need to be transitioned to desired levels to avoid imposing unacceptable price shocks on customers.
	Equitable	
	Certainty	
Equitable recovery of common network costs		Allocate common costs of network service provision to tariff classes on the basis of user pays principle to ensure that prices lie between avoidable and standalone cost.

The pricing objectives identified in the above table are discussed in more detail below:

Achieve economic efficiency

EnergyAustralia believes that it is important that network tariffs provide customers with price signals reflective of the economic cost of network service provision. This approach to setting network tariffs ensures that customers make economically rational decisions about their network use.

A key strategy for achieving economic efficiency is to set prices for the use of the electricity distribution network during periods when the network is more likely to be congested reflective of marginal cost. This approach ensures that customers are informed of the economic cost of their network use and are more likely to make economically rational decisions in relation to changing their behaviour or investing in more energy efficient household appliances.

While some customers may place a high value on their peak network use and, therefore, be willing to pay the higher price of network use during peak periods, other customers are more likely to respond to these peak price signals by minimising their use of network during periods of network congestion, which is likely to contribute to a reduction in peak demand and consequently less need to invest in network peak capacity.

A full discussion on economic network pricing concepts is found in chapter 3 of Part 2 of this document.

Achieve regulatory outcomes

To ensure the long-term viability of the network, regulators establish the annual revenue requirement of the distribution business by considering the efficient costs of owning, operating, maintaining and investing in the network over the regulatory period. This distribution revenue requirement is translated into X factors that cap the movement in weighted average price in each year of the regulatory period. This type of price control is referred to as a price cap.⁵

Under a price cap, the distributor is exposed to the risk that its tariffs will not generate sufficient distribution revenue to cover the efficient cost of owning, operating, maintaining and augmenting its distribution network. Therefore, an important aspect of the network price setting process is to develop strategies to deliver regulatory outcome in a manner that does not expose the business to unacceptable levels of distribution revenue risk. In an economic and volume environment of considerable uncertainty, this represents a significant challenge as it is important that revenue risk mitigation is achieved without compromising other pricing objectives, such as equity and efficiency.

Make prices comprehensible

Ideally, all customers should be able to understand how prices are determined and how their bills are calculated. Unfortunately, simplicity is not always consistent with achieving other goals. Developing prices which are as

equitable as possible, which encourage economic behaviour and which ensure the right return on investment would lead to quite complex price structures. How the actual prices are calculated for each charging parameter (eg network access charge) is also similarly complicated given the competing objectives underpinning the price setting process. A major aim of this document is to provide a broad understanding of the process used to develop network prices.

More complex network prices have tended to apply to larger customers such as businesses, while domestic and small business customers – who comprise over 90% of the total customer base – have had fairly simple price structures. However, to continue to provide small customers with a highly averaged pricing signal is not desirable on economic and equity grounds. Over the remainder of the regulatory period, EnergyAustralia will focus on exploring a range of reform options that aim to lay the foundation for more cost reflective pricing signals for all network customers. The challenge will be to develop pricing structures that are not only more cost reflective, but are also able to be understood by retailers and their end-customers.

Provide certainty over future price trends

It is not desirable to make unexpected changes to the structure and level of network charges as this is likely to undermine the willingness of customers to make permanent changes in their network use, such as investing in more energy efficient appliances given that it creates uncertainty over the extent to which these changes result in lower network bills in the future.

It is for this reason that EnergyAustralia is committed to providing external stakeholders with as much certainty as possible over the future direction of network prices. This commitment takes many forms, such as publishing the pricing proposal and statement of expected price trends, stakeholder engagement and consultation that aim to provide interested parties with an understanding of EnergyAustralia's approach to setting network tariffs and potential tariff reform options being investigated.

This commitment also directly influences the setting of network tariffs to ensure that prices are transitioned to cost reflective levels, where necessary to avoid imposing unexpected price shocks on customers. EnergyAustralia is confident that this approach will realise the benefits of more efficient prices without raising undue equity concerns.

Make prices equitable for all customers

EnergyAustralia seeks to ensure that network prices are equitable as well economically efficient. This concept of equity means that two customers with similar network usage patterns located next door to each other should expect to be charged on the same basis, customers whose location or consumption pattern costs the DNSP more to service should expect to pay a greater cost.

⁵ Unlike distribution, the recovery of transmission costs is governed by an overs and unders account ie revenue cap.

Equitable recovery of common network costs

EnergyAustralia believes that the allocation of common network costs to individual tariff classes on the basis of the 'user pays' principle is the most equitable approach to establishing how much distribution revenue to recover from each tariff class. In recent years, EnergyAustralia has been refining its cost allocation approach so that the amount of distribution revenue recovered from a particular tariff class is as close as possible to the cost incurred by EnergyAustralia to

provide network service to these customers. This approach will also ensure that distribution prices are likely to lie between avoidable and standalone cost, as required by the National Electricity Rules (as discussed in detail in Part 2 of this document). This is important as it means that network tariff setting process will not result in economic subsidies between tariff classes.

The following table provides a comparison of the allocated network cost and the forecast network revenue at the individual tariff class level.

Table 5: Cost Allocation Vs Network Revenue By Tariff Class – FY11

Tariff Class	Network Cost Allocation(\$m)	Forecast Network Revenue (\$m)	% Variation
Unmetered	9.8	13.5	37.8%
Residential	933.2	894.9	-4.1%
Low Voltage Business	829.0	962.2	16.0%
High Voltage	67.9	67.4	-0.74%
Subtransmission	20.4	18.8	-7.8%
Cost Reflective Network Price	78.9	76.6	-2.9%

The key point to note from the above comparison is that most tariff classes are within a reasonable range of cost reflective levels. The only exception is the unmetered tariff class. While it is desirable in theory to set network tariffs that achieve perfect cost reflectivity at the tariff class level, in practice such an approach would result in unnecessary pricing uncertainty given that the cost allocation process is influenced by relative volume movements. In practice, it is preferable to balance the competing needs of price path certainty and the user pays principle by transitioning tariffs to cost reflective levels over a reasonable time period. For example, EnergyAustralia has adopted transitional TUOS arrangements to avoid imposing unacceptable bill impacts on customers from directly passing through TransGrid's transmission charges in FY11.⁶

Overview of network pricing strategies

EnergyAustralia has been at the forefront of network pricing in Australia with the introduction of a wide range of tariff reforms over the past five years, such as the Introduction of ToU pricing to a large number of small business and domestic customers; increases in peak energy and peak capacity charges toward economic cost, simplification of capacity charge to be on a rolling twelve month window, rather than on a fixed 24 month window with a complex reset arrangement and the introduction of capacity charges for business customers who consume as little as 40 MWh per year.

The following table provides a high-level understanding of the pricing strategies that promote the achievement of one or more of the pricing objectives discussed above.

⁶ Note that TransGrid significantly reformed the structure of transmission charges in FY11 by removing the peak and shoulder energy charges and increasing demand charges by around 150%.

Table 6: Summary of EnergyAustralia's Network Tariff Strategy

Objective		Strategy/Comment
Achieve economic efficiency by providing customers with prices reflective of economic cost		Network tariffs have been designed to reflect economic cost to the extent possible given existing tariff structures and metering constraints. EnergyAustralia is also committed to investigating more cost reflective forms of network pricing, such as dynamic tariffs and more innovative approaches to signalling economic cost, such as dynamic rebates and to increasing the number of customers on more cost reflective network tariffs, where it is economically and socially desirable to do so. It is for this reason that EnergyAustralia has given small domestic customers on a TOU tariff the option of switching to the standard tariff (refer to Attachment G).
Achieve regulatory entitlements		Network tariffs have been developed in accordance with the Weighted Average Price Cap constraint and price limits and the pricing principles set out in the National Electricity Rules (refer to Part 2 of this document). As a result of increased uncertainty about future volume movements, greater emphasis has been placed on increasing fixed and capacity charges where this is economically justified and socially desirable to do so.
Satisfy customers	Comprehensible	EnergyAustralia is committed to simplifying the network tariffs where appropriate. It is for this reason that several network tariffs have been made redundant from 1 July 2010 (refer to chapter 5 of Part 2 of this document)
	Fairness	EnergyAustralia is committed to developing network tariffs and assigning customers to network tariffs on a non-discriminatory basis (refer to Attachment G).
	Stable	EnergyAustralia is committed to provide retailers and end-customers with certainty over the direction of network prices through publishing pricing documents, the retailer's forum and through the establishment of transitional price path arrangements to cost reflective levels.
Equitable recovery of common network costs		EnergyAustralia is committed to improving its allocation of network costs to individual tariff classes to ensure that the amount of distribution revenue recovered from customers in a particular tariff class is as close as possible to the costs that EnergyAustralia incurs on these customers' behalf.

EnergyAustralia believes that it is important to provide external stakeholders with an understanding of not only the basis of the proposed network prices for FY11, but also the potential medium to longer term direction of EnergyAustralia's network prices. In this way, retailers and other interested parties will be better aware of the potential reforms to network tariff structures and weightings between charging parameters. The following discussion provides an understanding of the key pricing initiatives that are most likely to influence the future direction of EnergyAustralia's network prices:

EnergyAustralia's Time of Use (ToU) electricity pricing program

EnergyAustralia has been at the forefront of Time of Use (ToU) metering and pricing innovation in Australia since 2004 with its ToU program, as summarised below:

- Disc meters at the majority of medium sized customer premises (in the consumption range of 15- 160 MWh per annum) have been replaced by interval meters. This meter replacement program has reached completion;

- Many customers with interval meters have been transferred to ToU prices; and
- All new and upgraded customer connections receive an interval meter. This includes new domestic dwellings as well as businesses.

The Independent Pricing and Regulatory Tribunal (IPART), in its 2004 Determinations, facilitated this ToU program by allowing expenditure for meter replacement and modifying the pricing side constraints to allow ToU prices to be mandated.

EnergyAustralia has rolled out interval meters to around 260,000 customers. By June 2014, it is anticipated that in the absence of a mandated Advanced Metering

Infrastructure role-out, over 500,000 customers will have an interval meter installed in their premise, making it the largest deployment of interval meters in Australia.

EnergyAustralia plans to undertake a major review of its ToU pricing program in FY11.

EnergyAustralia's strategic pricing study

EnergyAustralia has completed a Strategic Pricing Study (SPS) that was designed to quantify how customers would respond to tariff reform initiatives such as Seasonal ToU and Dynamic Peak Pricing.

A summary of the key results are shown in the following table.

Table 7: Residential Response Levels to Dynamic Tariffs

Tariff	Average across all DPP events	Extreme Temperatures		Assessment
		Winter (5°C)	Summer (31°C)	
Information Only	11%	11%	13%	Inconsistent, ranges from -1% to 24%
DPP Medium with IHD	25%	30%	36%	No difference in response between medium \$1/kWh and high \$2/kWh
DPP High with IHD	25%			
DPP High without IHD	23%			IHDs add minimal response

The key learnings from the study are:

- Domestic customers who sign up to the tariff respond well to dynamic tariffs, and are willing to reduce their air conditioning usage on hot days, despite popular belief to the contrary. The further away from the comfortable 18° to 22° Celsius range, the greater the demand response.
- Despite the apparent complexity of dynamic tariffs, customers are able to understand the concept.
- The dynamic price could probably be set at 50-70c/kWh and still achieve comparable reductions. The demand response to prices above \$1/kWh appears to be saturated.
- 0-160MWh p.a. business customers showed no response to dynamic tariffs during the trial. It may be that they would respond to price signals over the longer term.

While these results demonstrate that innovative network tariffs have the potential to play a role in demand management, more work is required to better

understand the equity, economic, financial and regulatory implications of these tariffs.

Proposed new tariff trial during 2009-14 Regulatory Period

EnergyAustralia is planning to build on the results of the Strategic Pricing Study by launching a new tariff trial during the current regulatory period. The trial will examine rebate tariffs as a method of achieving demand response. It is anticipated that this kind of tariff will have a much greater rate of customer sign-up, improving the expected aggregated demand response. The rebate arrangement is also operationally much simpler to implement, vastly improving its potential as a real tool for deployment within the next five years. The rebate tariff works by paying customers for load reductions. The more they reduce their usage, the more they are paid.

Compliance with regulatory constraints

The AER's 2009-14 Determination specifies a Weighted Average Price Cap (WAPC) control mechanism for standard (distribution) control services. The WAPC limits price movements over the regulatory period, so that the business receives revenue gains or losses depending if volumes rise or fall. The WAPC works by weighting each

proposed price by its historic volumes, and dividing this result by existing prices weighted by the same historic volumes. This “weighted average” price increase is limited to a combination of CPI, X factor and D factor for that year:

Weighted Average Price Constraint

The weighted average price cap formula is defined below:

$$\frac{\sum_{i=1}^m \sum_{j=1}^n p_t^{ij} \times q_{t-2}^{ij}}{\sum_{i=1}^m \sum_{j=1}^n p_{t-1}^{ij} \times q_{t-2}^{ij}} \leq (1 + \Delta CPI)(1 - X_t)(1 + D_t)$$

where:

- p_t = the proposed prices for the upcoming financial year
- p_{t-1} = prices for the current financial year
- q_{t-2} = volume quantities from the most recently completed financial year
- X_t = the allowed real average price increase set by the AER from their April 2009 Determination
- D_t = the AER approved adjustment to recover costs from successful demand management initiatives
- ΔCPI = the most recent historic average December to December change in CPI

For FY11, the change in CPI is 1.82%, the allowed X-factor is 18.18% and the D-factor is 0.1%. Distribution prices are therefore allowed to increase on a weighted average basis by around 21% for 2011.

Refer to Part 2 of this document for a comparison of the overall WAPC constraint and the weighted average distribution price increase for FY11.

Price Limits

In addition to the weighted average price constraint, increases to EnergyAustralia’s distribution prices are also constrained by price limits that apply on a tariff class,

rather than individual tariff basis. Specifically, the National Electricity Rules require that the expected weighted average distribution revenue to be raised from a tariff class must not exceed the overall CPI+X constraint plus 2%.

For FY11, the change in CPI is 1.82%, the allowed X-factor is 18.18%, the D-factor is 0.1% and the L-factor =2%. Distribution prices are therefore allowed to increase on a weighted average basis for each tariff class by just under 23% for 2011. Refer to page X for a comparison at the tariff class level of the calculated price limit and the weighted average distribution price increase for FY11.

4. Summary of Proposed Network Tariffs for FY11

This section aims to provide external stakeholders with an understanding of the proposed changes to the major published network tariffs (inclusive of climate change fund contribution) for FY11. Please note that the prices presented below have been rounded to two decimal places for presentation purposes. (please refer to chapter 3 of Part 2 of this document for a complete

listing of EnergyAustralia's proposed prices for FY11 for all of the published network tariffs).

The following table provides a comparison of the network tariffs (inclusive of climate change fund contribution) for FY10 and the proposed network tariffs for FY11 for residential customers.

Table 8: Proposed Network Tariffs – FY11 – Residential Customers

Charging Parameter	Actual FY10	Proposed FY11	% Variation
Inclining Block Tariff (EA10)			
Fixed Charges (C/day)	19.88	23.31	17.3%
First Block (C/kWh)	7.47	8.89	19.0%
Second Block (C/kWh)	10.49	13.80	31.6%
Time Of Use Tariff (EA25)			
Fixed Charges (C/day)	27.05	32.46	20.0%
Peak (C/kWh)	16.81	20.20	20.1%
Shoulder (C/kWh)	3.75	4.32	15.2%
Off-peak (C/kWh)	1.56	2.03	30.0%
Note – Prices have been rounded to two decimal places for presentation purposes. Exclude GST			

The following table provides a comparison of the actual network tariffs for FY10 and the proposed network tariffs for FY11 for controlled load customers.

Table 9: Proposed Network Tariffs – FY11 – Controlled Load Customers

Charging Parameter	Actual FY10	Proposed FY11	% Variation
Controlled Load 1 Tariff (EA30)			
Fixed Charges (C/day)	1.51	1.54	1.8%
Non-TOU Charge (C/kWh)	0.66	1.56	137.5%
Controlled Load 2 Tariff (EA40)			
Fixed Charges (C/day)	5.84	5.94	1.8%
Non-TOU Charge (C/kWh)	2.99	3.84	28.3%

The following table provides a comparison of the actual network tariffs for FY10 and the proposed network tariffs for FY11 for small business customers.

Table 10: Proposed Network Tariffs – FY11 – Small Business Customers

Charging Parameter	Actual FY10	Proposed FY11	% Variation
Inclining Block Tariff (EA50)			
Fixed Charges (C/day)	59.10	71.82	21.5%
First Block (C/kWh)	6.37	7.64	20.0%
Second Block (C/kWh)	9.55	11.57	21.2%
Time Of Use Tariff (EA225)			
Fixed Charges (C/day)	27.04	32.00	18.3%
Peak (C/kWh)	17.48	20.65	18.1%
Shoulder (C/kWh)	3.89	4.35	11.8%
Off-peak (C/kWh)	1.79	1.89	5.6%
Note – Prices have been rounded to two decimal places for presentation purposes. Exclude GST.			

The following table provides a comparison of the actual network tariffs for FY10 and the proposed network tariffs for FY11 for medium business customers.

Table 11: Proposed Network Tariffs – FY11 – Medium Businesses

Charging Parameter	Actual FY10	Proposed FY11	% Variation
Time Of Use Tariff 40MWh to 160 MWh pa (EA302)			
Fixed Charges (C/day)	71.83	100.81	40.3%
Peak (C/kWh)	7.67	9.83	28.1%
Shoulder (C/kWh)	5.65	6.91	22.3%
Off-peak (C/kWh)	1.94	2.45	26.1%
Capacity (c/kW/day)	11.65	13.91	19.4%
Time Of Use Tariff 160MWh to 750 MWh pa (EA305)			
Fixed Charges (C/day)	133.32	187.89	40.9%
Peak (C/kWh)	8.00	9.18	14.7%
Shoulder (C/kWh)	6.03	6.77	12.3%
Off-peak (C/kWh)	1.69	2.48	46.7%
Capacity (c/kVA/day)	11.49	14.50	26.2%
Note – Prices have been rounded to two decimal places for presentation purposes. Excluded GST			

The following table provides a comparison of the actual network tariffs for FY10 and the proposed network tariffs for FY11 for large business customers.

Table 12: Proposed Network Tariffs – FY11 – Large Businesses

Charging Parameter	Actual FY10	Proposed FY11	% Variation
Time Of Use Tariff Greater than 160 MWh pa (EA310)			
Fixed Charges (C/day)	227.64	320.82	40.9%
Peak (C/kWh)	7.10	7.73	8.9%
Shoulder (C/kWh)	5.58	6.63	18.8%
Off-peak (C/kWh)	1.49	2.40	61.1%
Capacity (c/kVA/day)	12.68	15.86	25.1%
High Voltage Time Of Use Capacity Tariff (EA370)			
Fixed Charges (C/day)	635.80	891.06	40.1%
Peak (C/kWh)	5.78	6.79	17.7%
Shoulder (C/kWh)	4.31	4.60	6.7%%
Off-peak (C/kWh)	1.03	2.05	99.0%
Capacity (c/kVA/day)	8.89	11.12	25.1%
Subtransmission Voltage Time Of Use Capacity Tariff (EA390)			
Fixed Charges (C/day)	732.1	1,025.97	40.1%
Peak (C/kWh)	2.76	3.26	18.4%
Shoulder (C/kWh)	2.10	2.49	18.7%
Off-peak (C/kWh)	0.67	1.08	63.8%
Capacity (c/kVA/day)	3.18	3.25	2.2%
Note – Prices have been rounded to two decimal places for presentation purposes. Exclude GST.			

5. Customer Impact Analysis

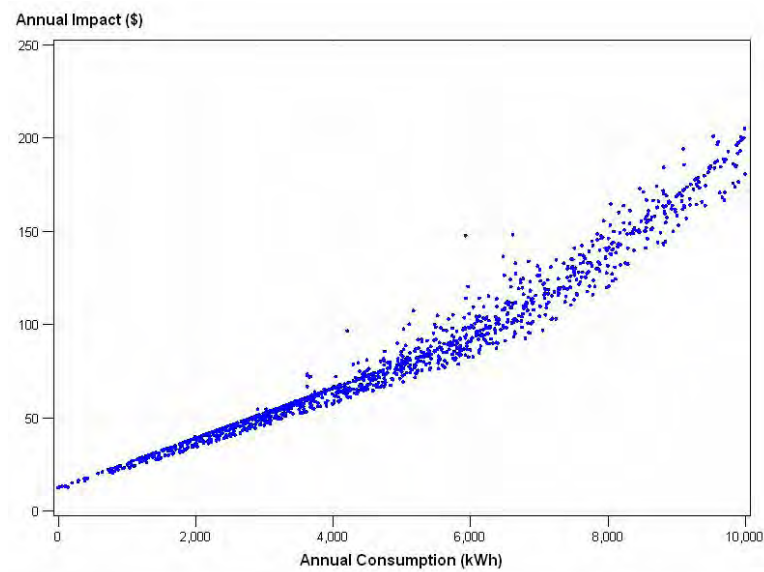
This section aims to provide external stakeholders with an understanding of the estimated annual network bill impacts of EnergyAustralia's major network tariffs for FY11.

The estimated network bill impact associated with the proposed transfer of customers from their existing network tariff to a more cost reflective network tariff in FY11 has also been provided. The estimated impact of transferring from inclining block tariffs to time of use tariffs is also provided.

Inclining Block Tariff (EA10) - Residential Customers

The following figure shows the estimated network bill impact for customers on the residential inclining block tariff.

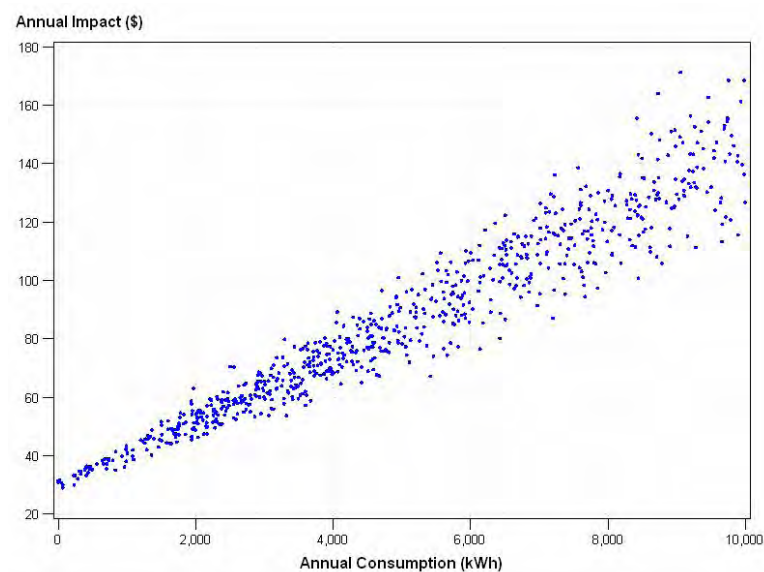
Figure 5: Network Bill Impact – Residential IBT – FY11



Time of Use Tariff (EA25) - Residential Customers

The following figure shows the estimated network bill impact for customers on the residential time of use tariff.

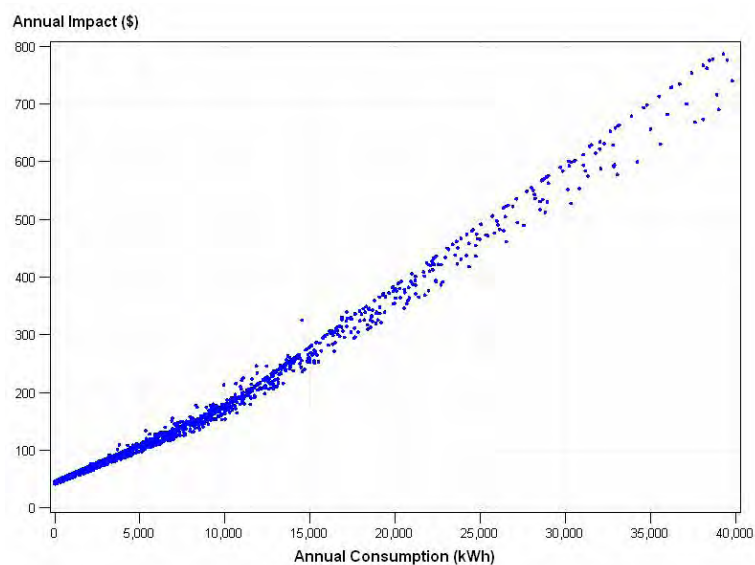
Figure 6: Network Bill Impact – Residential TOU – FY11



Inclining Block Tariff (EA50) – Small Business Customers (<40 MWh pa)

The following figure shows the estimated network bill impact for customers on the small business inclining block tariff.

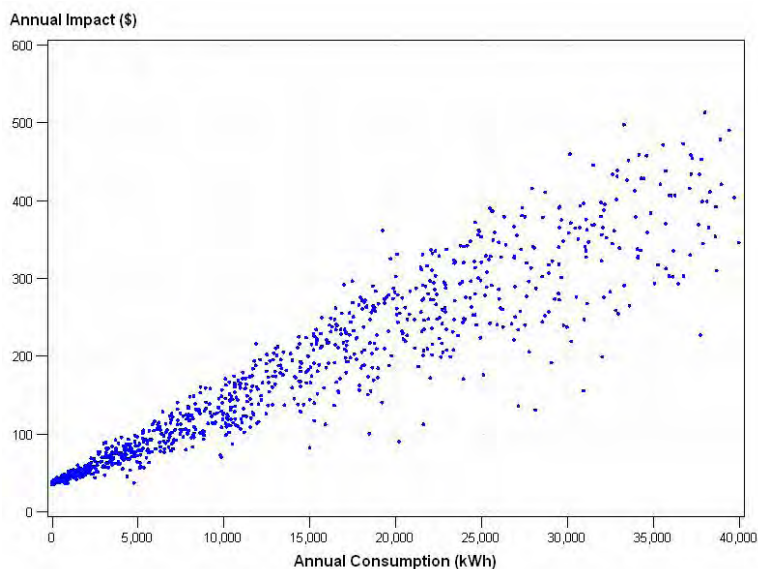
Figure 7: Network Bill Impact – Small Business IBT – FY11



Time of Use Tariff (EA225) - Small Business Customers (<40 MWh pa)

The following figure shows the estimated network bill impact for customers on the small business time of use tariff.

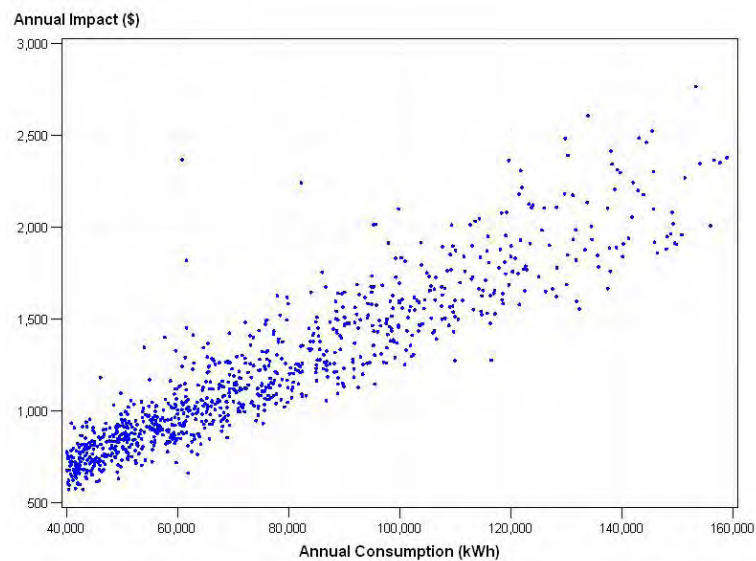
Figure 8: Network Bill Impact – Small Business TOU – FY11



Time of Use Capacity Tariff (EA302) – Business Customers (40 - 160 MWh pa)

The following figure shows the estimated network bill impact for customers on the medium-sized business time of use tariff.

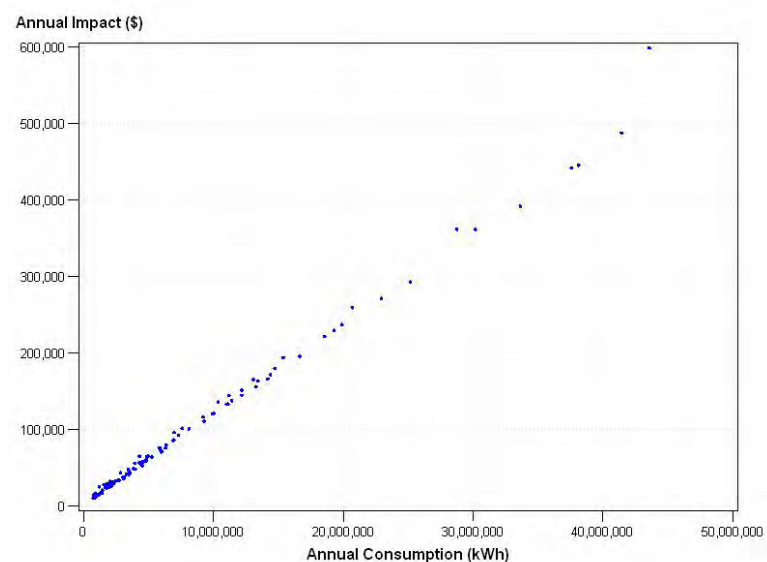
Figure 9: Network Bill Impact – Medium Business – FY11



Time of Use Capacity Tariff (EA370) – High Voltage Connected Customers

The following figure shows the estimated network bill impact for customers on the high voltage connection time of use tariff.

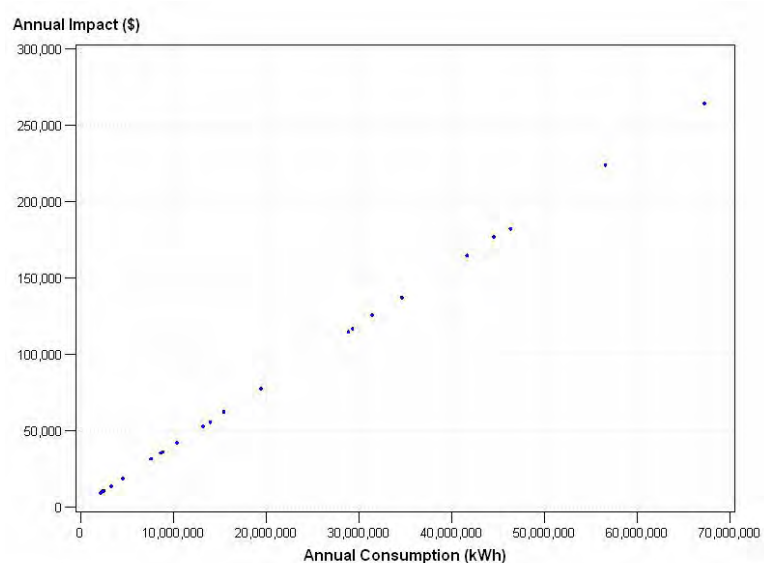
Figure 10: Network Bill Impact – High Voltage Connection – FY11



Time of Use Capacity Tariff (EA390) – Sub Transmission Voltage Connected Customers

The following figure shows the estimated network bill impact for customers on the subtransmission voltage connection time of use tariff.

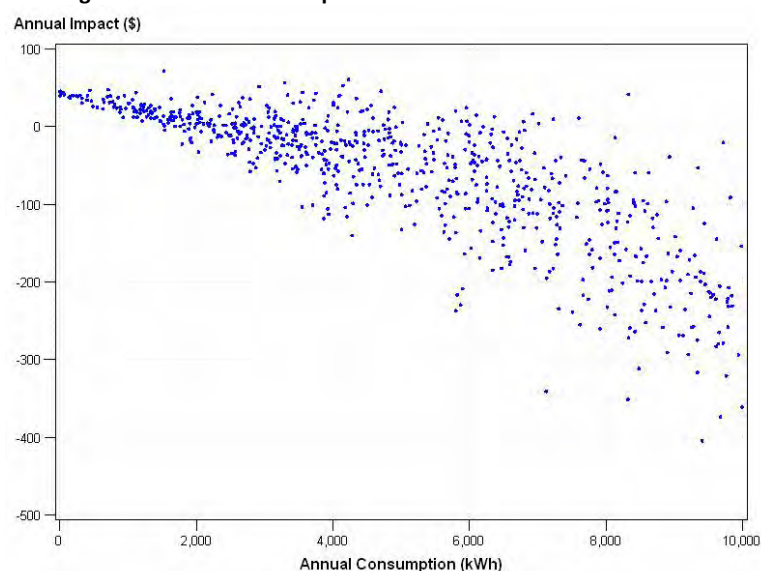
Figure 11: Network Bill Impact – Subtransmission Connection – FY11



Tariff transfer from Residential (EA010) to Residential (EA025) Time of Use Tariff

The following figure shows the estimated network bill impact for customers who are transferred from EA010 to EA025 during the course of FY11. Both prices used for comparison are FY11. Note: That negative dollar impact is a saving to the customer.

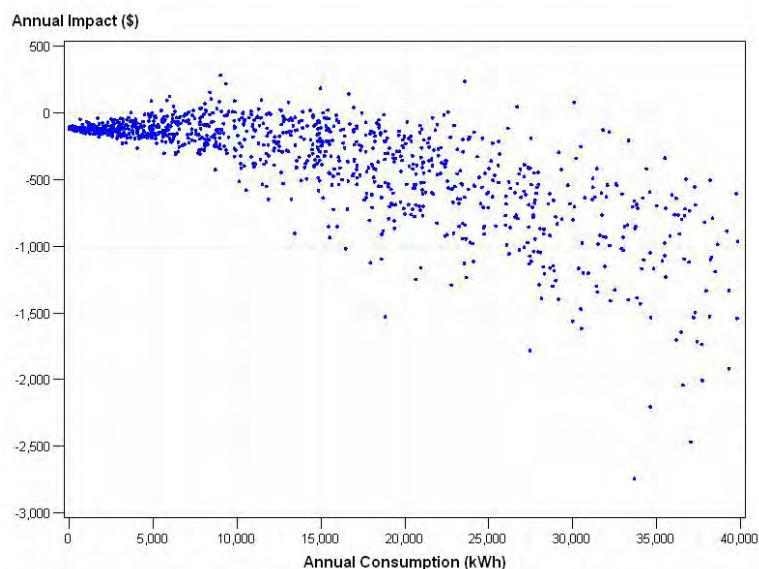
Figure 12: Network Bill Impact – Residential ToU transfers in FY11



Tariff transfer from Inclining Block Tariff (EA50) – Small Business Customers to Time of Use Tariff (EA225) - Small Business Customers

The following figure shows the estimated network bill impact for customers who are transferred from EA050 to EA225 during the course of FY11. Both prices used for comparison are FY11. Note: That negative dollar impact is a saving to the customer.

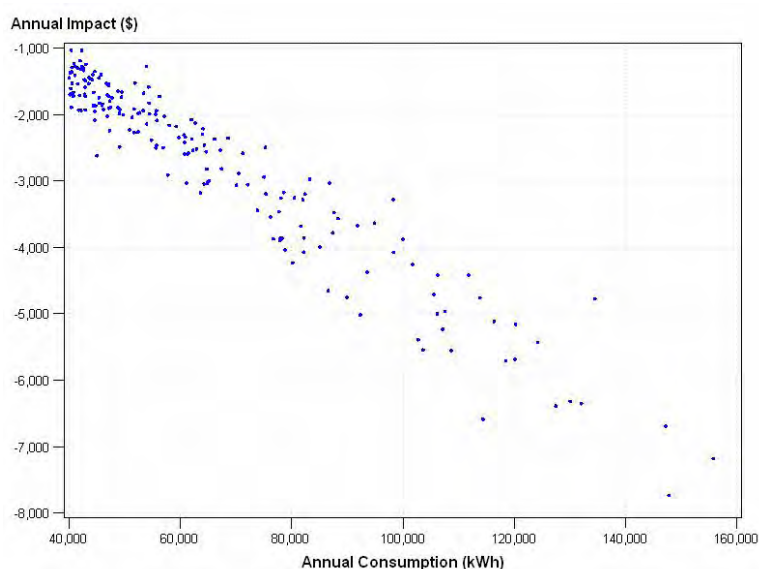
Figure 13: Network Bill Impact – Small Business ToU transfers in FY11



Tariff transfer from Inclining Block Tariff (EA50) – Small Business Customers to Time of Use Capacity Tariff (EA302) - Business Customers

The following figure shows the estimated network bill impact for customers who are transferred from EA050 to EA302 during the course of FY11. Both prices used for comparison are FY11. Note: That negative dollar impact is a saving to the customer.

Figure 14: Network Bill Impact – Small Business ToU transfers in FY11 (EA050 to EA302)



6. Glossary

Term	Definition
AER	Australian Energy Regulator
Bulk Supply Point (BSP)	A vital step in the transmission of electricity is decreasing the transmission line voltage using transformers. This is performed at a large substation called a Bulk Supply Point owned and operated in NSW by TransGrid
Capacity Charge	A capacity charge applies to the maximum half hourly kW or KVA recorded during the peak period of the day over the 12 months prior to a bill being calculated
Controlled load Tariff	Applicable to electricity which is separately metered and controlled by EnergyAustralia and used for operating storage water heaters, thermal storage space heaters, and other approved fixed wired appliances
Cost Reflective Network Prices (CRNP) Tariff	A price calculated for customers using more than 10MW, that takes into account load flows of the specific assets the customer uses
Distribution Use of System (DUOS)	The revenue earned for the provision of distribution services
Dynamic Peak Rebate	This is a rebate offered to customers for consumption reductions during a dynamic peak event
Dynamic Tariff	This tariff applies a price for energy consumed during a dynamically signalled congestion period
Economic Efficiency	The improvement in societal welfare from providing customers with a price signal reflective of marginal, rather than average cost of distribution service provision.
Equity	The impact of price changes on the well being of disadvantaged groups
First Block	See <i>Inclining Block Tariff (IBT)</i>
Fixed Charge	This tariff component does not vary with electricity usage at the customer's premises and is charged on a cents/day/NMI basis
Inclining Block Tariff (IBT)	This tariff typically has a fixed charge and a variable energy charge component with a low price charged up to a predetermined consumption threshold (first block) and a higher price charged to customers who consume electricity above the predetermined threshold (the second block)
Network Use of System (NUOS)	The revenue earned for the provision of transmission and distribution services
IPART	Independent Pricing and Regulatory Tribunal
Off-peak Consumption Charge	This tariff component applies to the price charged for electricity usage during the off-peak period of the day. See also <i>Time of Day Period Definition</i>
Overs & Unders Account (Transmission)	This account provides EnergyAustralia with certainty of transmission cost recovery by allowing future TUOS prices to increase (decrease) to recoup (return) an under (over)-recovery of TUOS
Peak Consumption Charge	This tariff component applies to the price charged for electricity usage during the peak period of the day. See also <i>Time of Day Period Definition</i>
Price Cap	Refer to Weighted Average Price Cap

Term	Definition
Real-time Tariff	This type of tariff has variable prices that vary every half-hour interval in response to change in energy purchase costs
Revenue Cap	A form of monopoly regulation where the regulated entity's allowed revenue from tariffs is limited by an X factor. Any over or under-recovered of actual revenue is accounted in an adjustment to tariffs in the subsequent period.
Revenue Requirement	The level of revenue that is adequate to fund the on-going operation of the business and provide a commercial return to the owner
Reasonable Estimates	A mechanism of the WAPC process that is used to adjust the historical volumes used in the formulae if customers are moved to another tariff. The intent of this reasonable estimate mechanism is to minimise gains or losses in revenue of moving customers between tariffs
Second Block	See <i>Inclining Block Tariff (IBT)</i>
Shoulder Consumption Charge	This tariff component applies to the price charged for electricity usage during the shoulder period of the day. See also <i>Time of Day Period Definition</i>
Time of Day Period Definition	The time periods of the day used to define when the peak, shoulder and off-peak price apply. The following is the current Time of Day Period Definitions: ➤ For Residential and Small Business up to 40 MWh per annum: ▪ Peak: 14:00 – 20:00 ▪ Shoulder: 07:00 – 14:00, 20:00-22:00 working weekdays, 07:00 – 22:00 weekends and public holidays ▪ Off Peak: All other times ➤ For all other ToU tariffs ▪ Peak: 14:00 – 20:00 ▪ Shoulder: 07:00 – 14:00, 20:00-22:00 working weekdays ▪ Off Peak: All other times
Time of Use (TOU) Tariff	This tariff typically has a fixed charge and a variable energy charge component with different prices applying at different time periods of the day. See also <i>Time of Day Period Definition</i>
Transmission Use Of System (TUOS)	The revenue earned from customers to recover the cost of transmission service provision by TransGrid and other <i>TNSPs</i>
Weighted Average Price Cap(WAPC)	A form of regulation that limits the annual increase to weighted average DUOS price to CPI+X. The weight used in the calculation of the average price is the previous year's volumes
X factor	The allowed rate of growth (positive or negative) in prices (under a price cap) or revenue (under a revenue cap)

ENERGYAUSTRALIA'S F11 NETWORK PRICING PROPOSAL

PART 2 – DEMONSTRATING COMPLIANCE

Compliance with national electricity
rules and the applicable distribution
determination

1. Overview and Outline

Part 2 of this document is EnergyAustralia's formal Pricing Proposal for FY11. This part of the proposal is submitted for review and approval by the Australian Energy Regulator (AER) as required by clause 6.18.2(a)(2) of the Transitional Chapter 6 Rules. It is structured to allow ready assessment of compliance by the AER.

Outline of compliance with Rules

EnergyAustralia's Pricing Proposal assesses all of the requirements set out in Rule 6.18.2 included in Chapter 11 (Transitional Rules) of the National Electricity Rules. Our pricing proposal also demonstrates compliance against the applicable distribution determination, as required under Clause 6.18.2(b)(7) of the Transitional Chapter 6 Rules.

Specifically, the proposal sets out the tariff classes, tariffs and charging parameters, and expected revenue for the second regulatory year of the new regulatory control period commencing 1 July 2010 and ending 30 June 2011.

This pricing proposal is structured as follows:

- Chapter 2 sets out the proposed tariff classes for standard control services, the basis for the proposed tariff classes, and EnergyAustralia's procedures for the assignment and reassignment of customers to tariff classes;
- Chapter 3 sets out the proposed tariffs and charging parameters, and the matters that EnergyAustralia has taken into account when determining these tariffs and charging parameters including the long run marginal costs of providing services to each tariff class, the transaction cost implications and the scope for customers to respond price signals;

- Chapter 4 sets out the weighted average revenue by tariff class and compares it to the efficient revenue boundary defined by the standalone cost and avoidable cost;
- Chapter 5 briefly discusses the variations to tariffs;
- Chapter 6 sets out the obligations on EnergyAustralia to provide revenue under the Climate Change Fund;
- Chapter 7 sets out the approach to passing through the cost of transmission use of system services;
- Chapter 8 sets out the changes since the previous regulatory year, as provided for in the Determination and the Rules; and provides information on how the pricing proposal complies with the requirements for pricing proposals as set out in the Rules.
- Chapter 9 sets out the tariffs and tariff class for EnergyAustralia's public lighting services for FY11.
- Chapter 10 sets out the proposed new network tariffs for FY11.

EnergyAustralia notes that Attachment D to this proposal demonstrates:

- Compliance with the WAPC control mechanism and side constraint formula
- Compliance with the application of reasonable estimates.
- Compliance with the control mechanism for EnergyAustralia's prescribed (transmission) standard control services.
- Compliance with the AER's requirements for reporting recovery of TUOS charges to account for under and over recovery of those charges.

2. Network Tariff Classes

RULE REQUIREMENT

Clause 6.18.2(b)(1) of the NER requires that a pricing proposal must set out the tariff classes that are to apply for the relevant regulatory year

In addition, when assigning customers to tariff classes the AER is required to have regard to the following principles;

(1) customers should be assigned to tariff classes on the basis of one or more of the following factors:

- (i) the nature and extent of their usage;*
- (ii) the nature of their connection to the network;*
- (iii) whether remotely-read interval metering or other similar metering technology has been installed at the customer's premises as a result of a regulatory obligation or requirement;*

(2) customers with a similar connection and usage profile should be treated on an equal basis;

(3) however, customers with micro-generation facilities should be treated no less favourably than customers without such facilities but with a similar load profile.

Clause 6.18.3(d) requires that a tariff class be constituted with regard to the need to group customers together on an economically efficient basis, and the need to avoid unnecessary transactions costs.

- Merging the low voltage residential tariff class and the low voltage business tariff class into a single low voltage tariff class;
- Assigning customers on a CRNP tariff to either the Sub-transmission voltage tariff class or the high voltage tariff class depending on their connection. The efficient pricing bounds test would only be applied at an individual CRNP customer level where a material risk of the customer by-passing the distribution network has been identified; and
- Merging the unmetered tariff class with the low voltage tariff class to form a single tariff class for all low voltage connected customers.

As a result, EnergyAustralia proposes to not change its tariff classes in FY11 and continue to assign customers to tariff classes on the following basis:

- the voltage connection required by the customer, namely low voltage (ie, 415V or 240V), high voltage (ie, 11kV or 33 kV) or sub-transmission voltage (ie 132/66/33kV);
- the usage of the customer (ie, maximum demand exceeding 10MW); and
- the metering installed in the customer's premise (ie metered or unmetered).

Table 1 defines EnergyAustralia's proposed tariff classes for direct control services in FY11.⁸ It should be noted that the proposed definitions of tariff classes encompass all customers that are provided with direct control services by EnergyAustralia, and are therefore required to be assigned to at least one of the specified tariff classes in accordance with the requirements of Clause 6.18.3(b) of the Transitional Rules.

Definition of a Network Tariff Classes

Under Chapter 10 of the NER, tariff classes are defined as representing 'a class of customers for one or more direct control services, who are subject to a particular tariff or particular tariffs'.

Proposed Network Tariff Classes

EnergyAustralia believes that there is scope to improve its definition of tariff classes⁷ to produce distribution pricing outcomes that better reflect the economic principles in the NER and improve the efficacy of the efficient pricing bounds test. EnergyAustralia intends to comprehensively review this area of our pricing proposal during FY11. The potential changes to how EnergyAustralia defines its tariff class under consideration are summarised below:

⁸ Please note that the tariff classes for Miscellaneous & Monopoly Services are set out in Attachment E. Tariff classes for public lighting services are set out in Chapter 9.

Table 1: EnergyAustralia's Proposed Tariff Class Descriptions - FY11

Tariff Class	Network Tariff	Definition
Residential Low Voltage	EA010 – Residential Non-TOU Tariff EA025– Residential Time of Use Tariff EA030 – Controlled load 1 EA040 – Controlled load 2	Applicable to separately metered low voltage (415V or 240V) connections.
Business Low Voltage	EA050– Business Non-TOU Tariff EA325 – LV Connection (Standby) EA050– Business Non-TOU Tariff EA225– Business Time of Use Tariff EA302 – LV 40-160 MWh (System) EA305– LV 160-750 MWh (System) EA310 – LV > 750 MWh (System)	Applicable to separately metered low voltage (415V or 240V) connections.
High Voltage	EA350 – HV Connection (Standby) EA370 HV Connection (System)	Applicable to any connection at high voltage (11kV) that is not otherwise eligible for a CRNP as defined below.
Sub-transmission	EA380 HV Connection (Substation)	Applicable to any connection at a sub-transmission voltage (132/66/33kV) that is not otherwise eligible for a CRNP price, as defined below.
Cost Reflective Network Prices (CRNP)	Customer Specific Prices	Applicable to connections at any voltage that use more than 10 MW of electricity demand on at least three occasions over a 12 month period.
Unmetered	EA401 Public Lighting EA402 Constant Unmetered EA403 Energylight	Applicable to any LV connection that is defined as an unmetered supply by EnergyAustralia in consultation with AEMO as per clause S7.2.3(Item 5) of the NER.

Procedures for the assignment and re-assignment of customers to tariff classes

EnergyAustralia will assign customers to tariff classes following the procedures set out in Appendix A of the final determination and by applying the following criteria:

1. a customer's site has a demand in excess of 10 MW per month on more than three occasions over a consecutive 12 month period, then regardless of the connection voltage, the customer is assigned to the CRNP tariff class for the purpose of demonstrating compliance with the NER at each annual price reset.
2. If the voltage of the supply is 33 kV, 66 kV or 132 kV then the customer is assigned to the

Subtransmission Voltage Capacity tariff class (unless already assigned to the CRNP tariff class).

3. If the voltage of the supply is 5 kV, 11 kV or 22 kV then the customer is assigned to the High Voltage Capacity tariff class (unless already assigned to the CRNP tariff class).
4. If the voltage of the supply is 415V or 240 V and the customer's annual consumption is below 40 MWh, the customer is assigned to either the residential or business Low Voltage tariff class depending on the nature of their usage.
5. If the supply is unmetered then the customer is assigned to the Unmetered tariff class.

The criteria for assigning customers to tariff classes is summarised in the following table.

Table 2: Criteria for Customer Assignment to Tariff Class

Tariff class	Nature of Network Connection	
	Voltage	Metering
Unmetered	415V/240V	Unmetered
Low Voltage Residential		Metered
Low Voltage Business		
High Voltage	5kV,11kV,22kV	
Sub-transmission Voltage	33kV,66kV,132kV	
CRNP (> 10 MW)	Any voltage	

If a customer's connection load or characteristics changes, the customer will be re-assigned to the appropriate tariff class. This will be carried out annually consistent with the annual review of customers' assignment to network tariffs (as discussed in Attachment G).

The review undertaken for this year has identified the need to re-assign a small number of customer sites from the Cost Reflective Network Price (CRNP) tariff class to either the Subtransmission or high voltage tariff class due to their maximum demand falling permanently below the 10 MW eligibility threshold to remain on a CRNP tariff.

It should be noted that EnergyAustralia intends to use reasonable estimates to account for customer transfers within a tariff class in FY11 under the WAPC, as summarised below:

- Around 37,500 residential and small business customers with type 5 meter installed in their premise from inclining block tariff to the appropriate TOU tariff.
- Around 5,300 customers with historical consumption of less than 40 MWh pa from the LV TOU Capacity Tariff (EA302) to either the Low Voltage Business TOU Tariff (EA225) or an appropriate Low Voltage TOU Capacity tariff, according to their historical annual consumption.
- Around 1,000 customers from the low voltage TOU Capacity tariff (EA305) to either the Low Voltage Business TOU Tariff (EA225) or an appropriate Low Voltage TOU Capacity tariff, according to their historical annual consumption.

- Around 1,000 customers from the low voltage TOU Capacity tariff (EA310) to either the Low Voltage Business TOU Tariff (EA225) or an appropriate Low Voltage TOU Capacity tariff, according to their historical annual consumption.

Customers with micro-generation facilities

It is EnergyAustralia policy to treat customers with micro-generation facilities no less favourably than customers without these facilities but with a similar consumption profile. Allocation of an embedded generation customer to a tariff class will be made on the same basis as other customers; this being the extent and nature of usage, and the nature of the connection to the network. The network tariff will include fixed and variable components and if the customer's demand were to be met entirely by the micro-generator then the levied charge will only be the fixed connection component.

EnergyAustralia's compliance with this requirement is further evident from the proposed network pricing arrangements for customers participating in the NSW Government Solar Bonus Scheme, which ensures that these customers are treated no less favourably than other customers as the billed consumption of these customers will be unaffected by their participation in the scheme. However, customers will be required to be a network TOU tariff to participate in the NSW Government Solar Bonus Scheme

3. Proposed Tariffs and Charging Parameters

RULE REQUIREMENT

Clause 6.18.2(b)(2) requires that the pricing proposal set out the proposed tariffs for each tariff class; and

Clause 6.18.2(b)(3) requires that the pricing proposal set out the charging parameters and the elements of service to which each charging parameter relates,

Rule 6.18.5 sets out the pricing principles that are relevant when determining tariffs and charging parameters.

Rule 6.18.5 further provides that:

(b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class:

(1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and

(2) must be determined having regard to:

(i) transaction costs associated with the tariff or each charging parameter; and

(ii) whether customers of the relevant tariff class are able or likely to respond to price signals.

(c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

This chapter sets out the proposed tariffs and charging parameters for direct control services for each tariff class for the period of July 2010 to June 2011, and explains the elements of service to which each charging parameter relates. The charging parameters for alternative control services (public lighting services) are discussed in Chapter 9 and set out in Attachments B and C. The charging parameters for Miscellaneous and Monopoly Services are in Attachment E.

The proposed tariffs and charging parameters are set out for each tariff class in Table 3. This chapter also explains how EnergyAustralia has applied the pricing principles set out in clause 6.18.5 of the NER. In developing the charging parameters, EnergyAustralia has taken account of a range of estimates of long run marginal cost, and has had regard to transactions costs and the scope for customers to respond to price signals. In addition, EnergyAustralia has ensured that its network tariffs are designed to recover the difference between marginal and average cost with the minimum of distortion to efficient patterns of consumption.

Table 3: EnergyAustralia's Proposed FY11 Network Tariffs by Charging Parameter

Tariff Class	Tariff Code	Tariff Name	Network Access Charge	Network Energy Rates				Step Rates (¢/kWh)		Daily Capacity Rates	
				Non-ToU (¢/kWh)	Time of Use (ToU) (¢/kWh)						
					Peak	Shoulder	Off Peak	Block 1	Block 2	(¢/kW/d)	(¢/kVA/d)
Domestic	EA010	LV Res non-TOU (Closed)	23.3146	—	—	—	—	8.8922	13.7961	—	—
	EA025	LV Res <40 MWh (System)	32.4600	—	20.1958	4.3206	2.0272	—	—	—	—
	EA030	Controlled Load 1	1.5406	1.5574	—	—	—	—	—	—	—
	EA040	Controlled Load 2	5.9444	3.8400	—	—	—	—	—	—	—
Low Voltage Business	EA050	LV Bus non-TOU (Closed)	71.8200	—	—	—	—	7.6448	11.5700	—	—
	EA225	LV Bus <40 MWh (System)	32.0000	—	20.6486	4.3483	1.8858	—	—	—	—
	EA302	LV 40-160 MWh (System)	100.8112	—	9.8264	6.9074	2.4469	—	—	13.9129	—
	EA305	LV 160-750 MWh (System)	187.8872	—	9.1752	6.7705	2.4796	—	—	—	14.5054
	EA310	LV >750 MWh (System)	320.8158	—	7.7288	6.6251	2.4031	—	—	—	15.8629
	EA325	LV Connection (Standby)	333.6024	—	6.1406	4.8594	1.1357	—	—	—	—
High Voltage	EA350	HV Connection (Standby)	928.5203	—	4.8594	3.6500	0.8754	—	—	—	—
	EA370	HV Connection (System)	891.0610	—	6.7931	4.6000	2.0500	—	—	—	11.1224
	EA380	HV Connection(Substation)	891.0610	—	6.0347	4.4832	1.1046	—	—	—	7.0291
Sub-trans VoltageCapacity	EA390	ST Connection	1025.9677	—	3.2560	2.4926	1.0808	—	—	—	3.2500
Unmetered	EA401	Public Lighting	—	6.0424	—	—	—	—	—	—	—
	EA402	Constant Unmetered	—	7.3939	—	—	—	—	—	—	—
	EA403	Energy-Light	—	5.4754	—	—	—	—	—	—	—

There are three elements of service for which charging parameters are defined, namely:

- network access charge per day per connection (€/connection/day) – this is a flat charge per connection, charged on the basis of the number of days of access to the network for that connection. A connection is defined as any connection to the network requiring a unique National Metering Identifier (NMI) as defined by the AEMO National Metering Identifier Procedure;
- electricity usage charge (€/kWh) – this is a charge per unit of electricity supplied to a connection point, through a primary tariff or also through an additional secondary tariff (usually controlled load hot water as seen from the Non-ToU charging parameter); and
- capacity charge per kilowatt or kiloVoltAmp, per day (€/kW/day or €/kVA/day) – this is a charge based on a customer's maximum

demand recorded in any half-hour period between 2pm and 8pm over the twelve months preceding the end of the current billing month. Maximum demand is recorded in either kW or kVA according to the customer's metering functionality.

In addition, there are a number of alternative components for the electricity usage charge including:

- an inclining block charge – this charge which is triggered once a threshold amount of electricity use has been used in each billing quarter. All electricity used after this point is charged at a higher rate based on (€/kWh). The threshold is 1,750kWh for domestic customers and 2,500 kWh for business customers.
- time of use consumption charges vary according to the time of day of consumption and whether it is a weekday, or weekend/public holiday, as summarised in Table 4.

Table 4: Time of Use periods for Tariffs

Customer Type	Annual consumption	Time of Use period definition
Residential and Small Business	< 40 MWh per annum	Peak period: 2.00 pm – 8.00 pm on working weekdays. Shoulder period: 7.00 am – 2.00 pm and 8.00 pm – 10.00 pm on working weekdays and from 7.00 pm – 10.00 pm on weekends and public holidays Off-Peak period : All other times
Medium and Large Business	> 40 MWh per annum	Peak period: 2.00 pm – 8.00 pm on working weekdays. Shoulder period: 7.00 am – 2.00 pm and 8.00 pm – 10.00 pm on working weekdays Off-Peak period : All other times

EnergyAustralia uses the Inclining Block Tariff (IBT) structure for customers mainly with Type 6 metering installations. The inclining block pricing structure is designed to encourage energy conservation, particularly for customers with air-conditioners who use significantly more energy (and contribute more to summer peak network congestion) than other customers. It also provides a financial incentive for large use customers to switch to a more cost reflective time of use tariff.

The price differential between the first and second block consumption under the IBT structure has been set based on an assessment of customers' capacity to respond to the price signal. Currently the second block price is around 50% higher at the NUOS level than the price charged for first block consumption.

As illustrated in the following figure (Figure 1), the IBT provides a strong financial incentive (ie higher network bills) to customers with consumption significantly second block consumption to change their consumption behaviour under the IBT or by voluntarily switching to the more cost reflective TOU tariff.

It is also relevant to note that customers that satisfy the criteria for EnergyAustralia's opt-out policy that choose to opt out of the TOU tariff will be placed on the inclining block tariff. This approach is consistent with EnergyAustralia commitment to pursuing network tariff reform in a socially responsible manner by striking an appropriate balance between economic and equity objectives.

It is important to note that EnergyAustralia proposes to investigate the potential economic efficiency benefits of further reform to the inclining block tariffs to ensure that customers on these tariffs continue to receive appropriate incentives to change their consumption behaviour or to switch over to the TOU tariff.

EnergyAustralia remains aware that any change to the price differential and consumption thresholds under the IBT structure has the potential to create material adjustment costs for some customers. Therefore the economic benefits of reform will need to be considered in the context of other considerations, such as equity and the preference of customers for simple tariffs. It is also important that external stakeholders are given adequate advance notice of reforms in this area and that reforms are implemented, if possible, in a transitional manner to give customers more time to adjust their consumption patterns to mitigate any adverse bill impacts arising from the reform process.

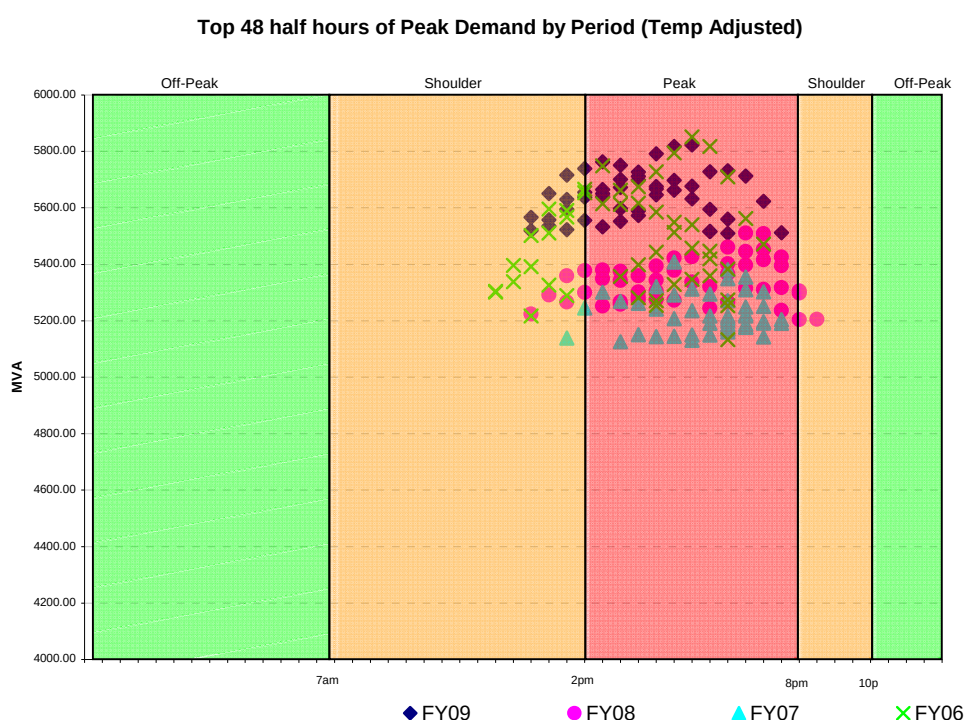
The current period definition for time of use (peak, shoulder and off-peak) is designed to cover both the evening winter and summer peak across the whole

network. EnergyAustralia believes that this broad definition, particularly for the peak period, has the potential to undermine economic efficiency even if peak energy charges are set reflective of LPMC.

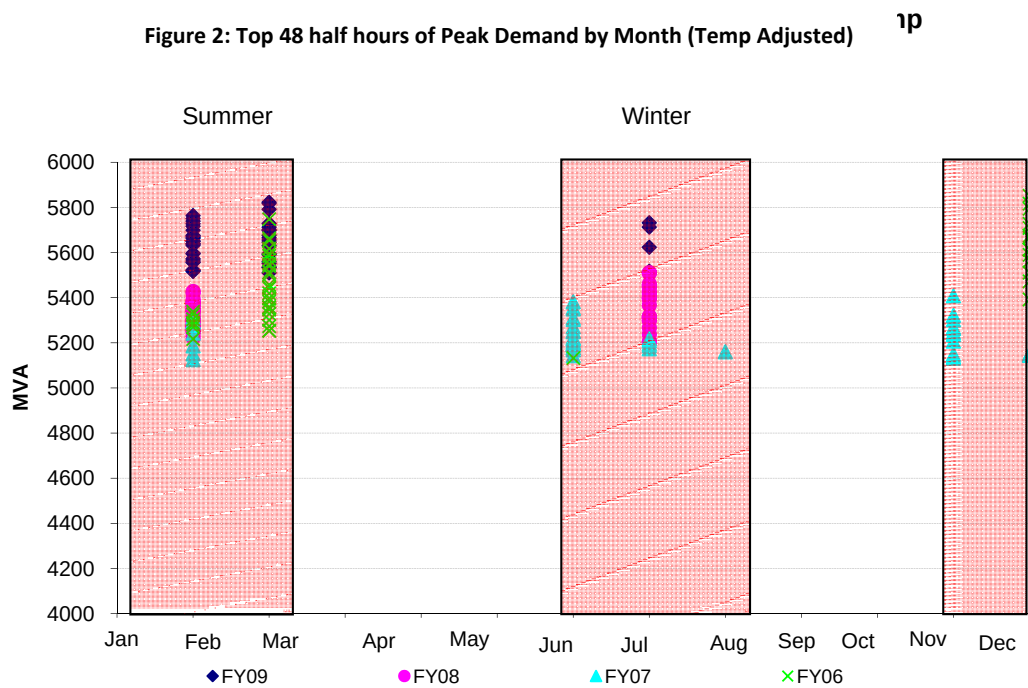
As with the potential IBT reforms, EnergyAustralia intends to investigate this matter and keep external stakeholders informed of developments in this area prior to a decision on whether or not to reform the TOU tariff structure in FY12. EnergyAustralia will also approach this reform in a manner that aims to maximise economic benefits without imposing material adjustment costs. EnergyAustralia believes that this approach to tariff reform will encourage permanent changes in consumption behaviour by providing customers (and their retailers) with greater certainty over the long-term direction of prices and pricing structures.

An analysis of the timing of the top 48 half hourly peak consumption intervals over recent years indicates that there are likely to be economic benefits to be realised from further reform to the peak period to ensure that peak price signals better reflect emerging patterns of network congestion.

Figure 1: Historical Pattern of Peak Demand – Half Hour Intervals



The following figure shows that the peak demand intervals occur in the warmer and cooler months of the year.



Further analysis indicates that the nature of peak network congestion is different in winter and summer, and is presented in Figure 3 and Figure 4 respectively.

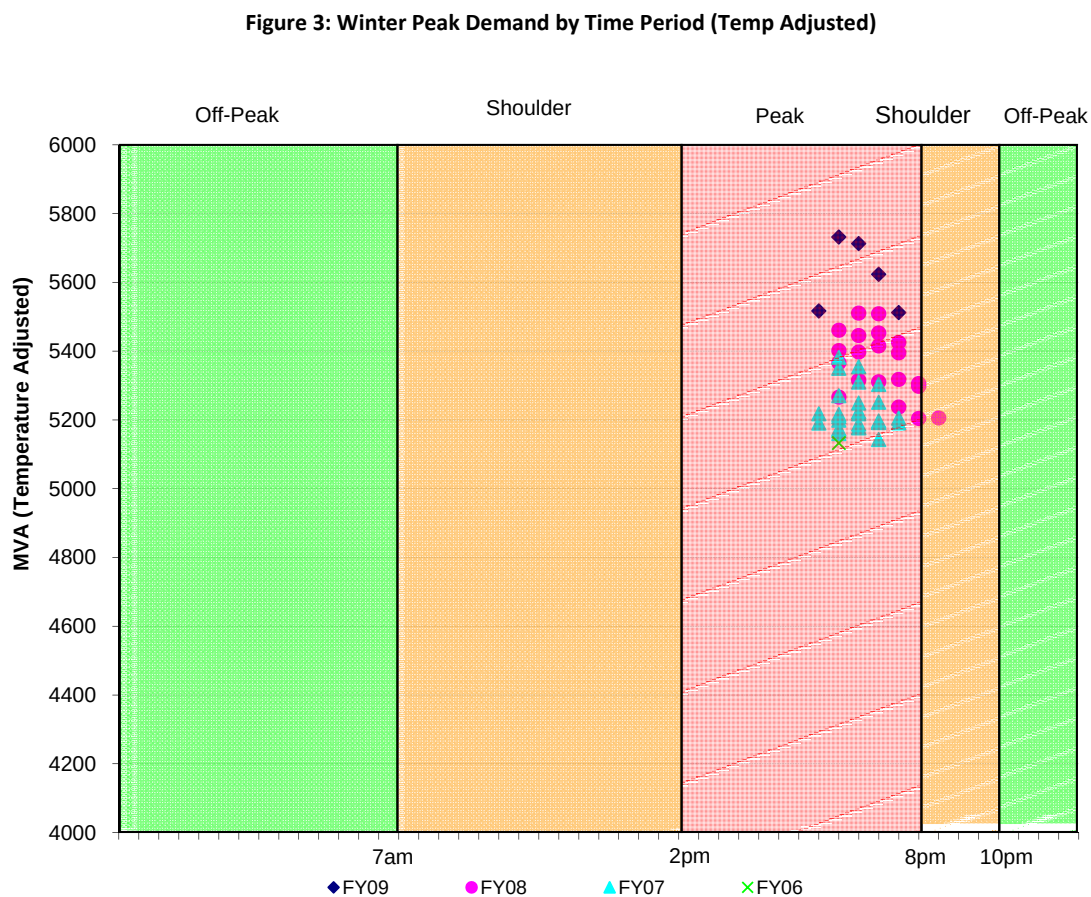
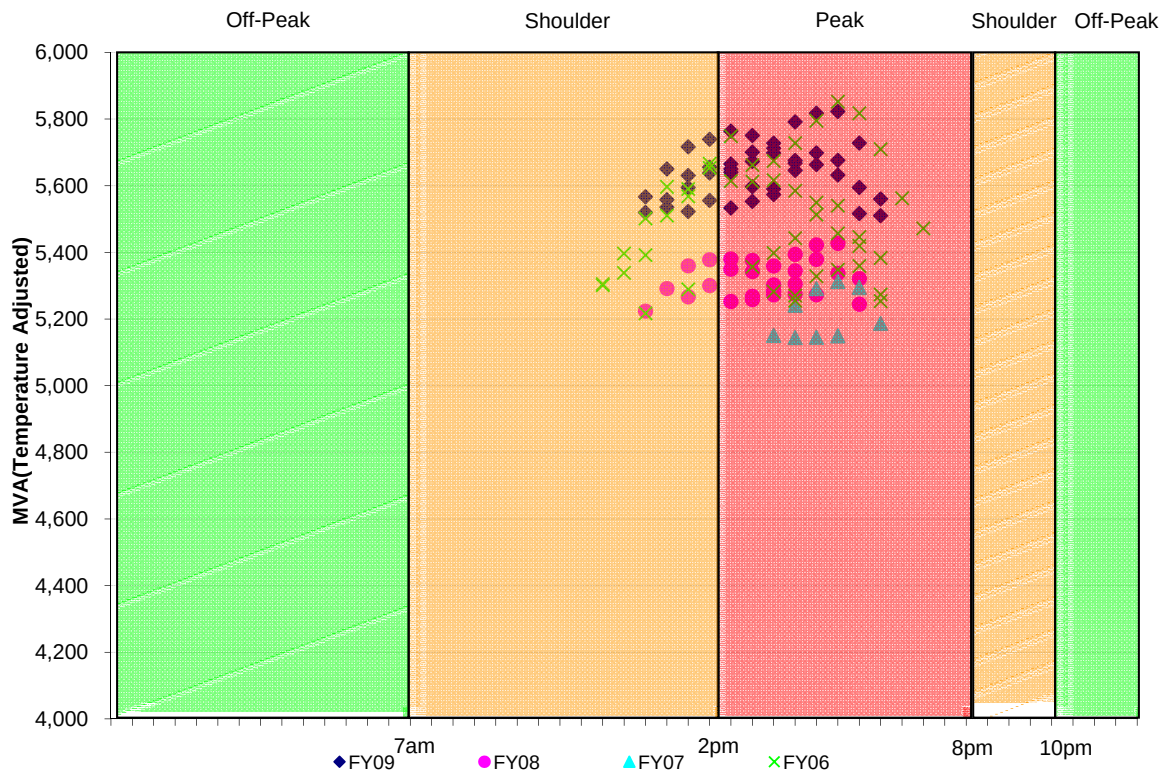


Figure 4: Summer Peak Demand by Time Period (Temp Adjusted)



Long run marginal costs

The concept of marginal costs is important in economics because of the general principle that setting prices equal to marginal costs presents consumers with the opportunity costs of their consumption decisions, and results in an efficient allocation of society's resources.

The timeframe over which marginal costs are considered to vary with consumption is the distinguishing difference between the concepts of short-run marginal cost (SRMC) and long-run marginal cost (LRMC). In the short-run only costs that directly vary with demand can be avoided (such as the need for some repairs and maintenance, or call centre costs), while in the long-run capital costs can be avoided particularly if anticipated demand growth does not eventuate. The LRMC is equal to the SRMC plus the marginal capital costs

The use of LRMC as a basis for determining usage charges is common in network businesses to provide customers with signals about the incremental capital and operating costs associated with use of the network. The use of LRMC is preferred over SRMC because of a desire to minimise the scope for constraints in the network to ration use of the network. In principle, the use of LRMC ensures that sufficient network capacity is made available such that demand is always able to be met by supply capacity.

Importantly, LRMC is a forward looking concept and should be estimated taking into account future expectations of the growth in costs to meet expected

increases in demand. There are a number of methods commonly used to estimate the LRMC for network businesses including the Average Incremental Cost (AIC) approach and the Turvey methodology.

These two approaches differ in the methodology used to estimate the incremental impact of changes in demand on capacity related expenditure. The AIC approach calculates the average of expected capacity related expenditure over the period that output is expected to change. In contrast the Turvey approach directly estimates the change in expenditure resulting from an increment or decrement in forecast demand.

While there are strong theoretical underpinnings in favour of the Turvey approach, EnergyAustralia has instead estimated the LRMC for each of the proposed tariff classes using the AIC approach. That said it intends to investigate the appropriateness of alternative approaches to estimating LRMC for tariff classes for future pricing proposals.

The AIC approach can be expressed algebraically as follows:

$$LRMC(AIC) = \frac{PV(capex) + PV(opex)}{PV(incremental_demand)}$$

where the optimal capital costs are the annualised capital expenditure required to meet the additional demand forecast over the period and the operating costs are those required to serve additional demand over the

forecast period. The additional demand served is the forecast change in demand compared to a base year. EnergyAustralia considers that incremental demand (rather than energy consumption) is the most appropriate volume cost driver for the LRM analysis.

The Determination sets out EnergyAustralia's optimal capital expenditure program to meet its overall demand

forecasts. This information has been used as the basis for determining the optimal capital costs to be allocated to each individual network tariff. Similarly the additional demand served has also been allocated to these network tariffs. The resulting range of estimates of the LRM were aggregated by proposed network tariff class as set out in the following table.

Table 5: EnergyAustralia's Long Run Marginal Cost

Tariff Class	LRMC \$ per KVA		
	Lower Bound of Estimate	Point Estimate	Upper Bound of Estimate
Low Voltage Residential	114.73	114.73	140.98
Low Voltage Business	-391.87	179.95	185.12
High Voltage	81.84	81.84	159.58
Sub-transmission Voltage	23.38	23.38	30.04
CRNP	-139.63	112.26	118.26
Unmetered	83.61	83.61	303.8

EnergyAustralia LRM model is extremely sensitive to key assumptions and inputs, as reflected in the broad range of LRM estimates shown for some tariff classes in the above table. EnergyAustralia believes that there is further scope to improve its estimates of LRM and remains committed to undertaking a major review in this area of the price setting function during FY11.

The application of LRM to the setting of efficient network tariffs is a two step process, as summarised below:

- To select the charging parameter to use to signal LRM to customers.
- To convert the LRM estimate from a \$ per kVA basis to a cents per kWh basis, if required.

EnergyAustralia believes it is efficient (and appropriate) that LRM estimates be reflected only in the setting of charging parameter(s) that cover the period when there is a material likelihood of system-wide network congestion, which are a key driver of network augmentation costs. A review of EnergyAustralia's existing charging parameters indicates that there are strong economic grounds to reflecting LRM in the setting of peak energy charges given that there is a reasonable probability that usage during peak periods will be correlated with network driving augmentation costs.⁹ It is important to note that future network tariff

reforms could result in the capacity charge, rather than peak energy charges, being the most efficient charging parameter to reflect LRM. The potential reform options for the capacity charging parameter will focus on improving the economic efficiency of the peak price signal by recognising that marginal costs of peak network capacity vary by the following dimensions:

- Time of year – the network is generally constrained during high peak demand periods of summer and winter
- Time of day – peak demand during the day tends to reflect temperature conditions and their impact on heating and cooling loads; and
- Location in the network – the balance between peak demand and network capacity tends to vary within the network area.

In recognition of the importance of LRM based pricing to customers and other external stakeholders, EnergyAustralia intends to review the methodology and modelling of LRM during FY11 to ensure that this important basis to the setting of peak charges is consistent with best practice. As with other areas of potential reform, EnergyAustralia will ensure that external stakeholders are informed of any changes to the calculation of LRM over the remainder of the current regulatory period and the methodological and/or modelling basis of these changes.

⁹ This correlation is likely to strengthen if the peak period definition is reformed to more accurately reflect summer and winter patterns of network congestion.

Given the decision to reflect LPMC in the peak energy charges, it is necessary to convert the LPMC estimates for each tariff class from a \$ per kVA basis to a cents per kWh basis. The formula used by EnergyAustralia to make this conversion is shown below:

$$LPMC = \frac{LPMC \text{ Estimate } (\$/kVA)}{\text{peak hours per annum} \times \text{power factor}}$$

Applying this formula results in the conversion of the LPMC estimate from a \$ per kVA to a cents per kWh basis, at each voltage level, as summarised in the table below:

Table 6: EnergyAustralia's Long Run Marginal Cost Estimate – c/kWh

Tariff Class	Annual Hours of Peak Period	Assumed Power Factor	Range of LPMC Estimate (Cents per kWh)		
			Lower Bound	Point Estimate	Upper Bound
Low Voltage Residential	1506	0.85	6.48	6.48	7.96
Low Voltage Business	1506	0.85	-22.12	10.16	10.45
High Voltage	1506	0.85	4.62	4.62	9.01
Sub-transmission Voltage	1506	0.85	1.32	1.32	1.70
CRNP	1506	0.85	-7.88	6.34	6.67

As required by the NER, EnergyAustralia has taken into account the estimated range of LPMC shown in the above table when setting the peak energy charging parameter of network tariffs for FY11. To achieve an efficient pricing outcome, the peak charge in DUOS terms has been set consistent with this LPMC range in FY11. This has meant that for some network tariffs, where the current peak charge is above or close to the upper range of LPMC, the increase to the peak charge has been minimised in DUOS terms in FY11. This has resulted in a larger DUOS increase being applied to the other charging parameters than would otherwise have been the case. As required under clause 6.18.5(c), this rebalancing of DUOS tariffs has been undertaken in a manner that minimises the distortion to efficient consumption patterns.

It is also important to note that EnergyAustralia has undertaken considerable rebalancing of TUOS tariffs in FY 2011 to ensure that the pursuit of more efficient DUOS pricing does not create unacceptable TUOS price trend uncertainty for network customers.

EnergyAustralia strongly believes that this is a reasonable approach to adopt considering the need to balance economic objectives with the need to customers with a reasonable degree of pricing certainty at the TUOS level.

Transactions costs

In developing the tariffs and charging parameters, EnergyAustralia has been mindful of minimising the transactions costs associated with levying charges. The

decision to charge on the basis of a combination of fixed, usage and capacity charges reflects the trade off between applying a complex set of charges (eg, introducing a power factor charge or additional inclining block rates) and ensuring that charges can be closely aligned to the underlying marginal costs of providing direct control services.

Introducing a greater number of tariff and charging parameters increases pricing complexity and will inevitably increase the transactions costs involved in charging for network services, both to EnergyAustralia through the need for more complex (and expensive) customer management and billing systems, and to customers, through additional effort required to understand the basis upon which charges are calculated.

EnergyAustralia believes that its network tariffs and charging parameters strike the right balance between transactions costs and ensuring that appropriate signals are provided to customers to facilitate the efficient use of network services.

Scope to respond to price signals

EnergyAustralia's network tariffs and charging parameters provide both short term and long term incentives to customers to modify consumption patterns in line with efficient use of network services. These price signals are conveyed to consumers of network services through the use of energy and/or capacity charges, depending on metering constraints and have been designed with the principal aim of encouraging customers to either reduce their usage of the network

during the peak period. Short term changes to consumption can include moving electricity usage out of the time-of-use peak periods. Long term changes typically occur with installation of, more effective insulation for housing, more energy efficient plant and equipment, or the installation of power factor correction devices e.g. capacitor banks.

EnergyAustralia has undertaken a trial to test residential and business customer demand response to new tariff initiatives. The Strategic Pricing Study (SPS) was a two year study that was completed in August 2008 (refer to part 1 of this document for more details on the SPS and EnergyAustralia's TOU program).

The results of the trial were encouraging. Over the 13 events that were called during the two year period, domestic customers showed energy reductions of up to 30% (winter) and 36% (summer), when compared to the control group.

EnergyAustralia is continuing to explore the potential of such tariffs during the 2009-14 regulatory period. We intend to develop a commercial offering if such tariffs can be demonstrated to be an effective demand management tool.

4: Weighted Average Revenue

RULE REQUIREMENT

Clause 6.18.2(b)(4) of the NER requires that a pricing proposal set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year.

This chapter sets out the expected weighted average revenue from tariffs within each tariff class for standard control services. In addition, it compares these revenues

with the standalone costs and avoidable costs by tariff class to demonstrate that the proposed tariffs comply with the principles set out in the NER.

The weighted average revenue for FY10 and FY11, both including and excluding government levies, is set out in the following table.

Table 7: Weighted Average DUOS Revenue by Tariff Class (\$ million excluding GST)

Tariff Class	Weighted average distribution revenue FY10 (\$m)	Weighted average distribution revenue FY11 (\$m)
Domestic	\$617.9	\$744.6
LV Business	\$560.0	\$672.4
HV Business	\$35.6	\$38.2
Sub-transmission	\$4.7	\$5.3
CRNP	\$30.1	\$33.4
Unmetered	\$7.1	\$9.9

Estimating standalone costs

The standalone costs represent the costs that would be incurred to replicate or bypass the infrastructure used to provide a service. It represents an upper bound of costs that should be recovered from customers. If customers were paying above the standalone costs then it follows that it would be economically beneficial for customers to switch to an alternative provider, and similarly if entry was economically feasible it would result in new suppliers entering to provide services. This creates the possibility of inefficient bypass of the infrastructure.

To estimate the standalone costs for a tariff class, EnergyAustralia has used the construct of a hypothetical new entrant distribution network service provider that is seeking to supply services to each tariff class separately. EnergyAustralia has then asked what infrastructure and operating costs would be incurred to supply the tariff class alone, without the benefit of any economies of scale and scope arising from the supply of services to all other tariff classes. To assist with estimating the cost of the hypothetical new entrant, EnergyAustralia has drawn upon information from its own costs as follows:

- Identification of the segments of the network that would be required to provide services to each tariff class. For example, for HV Business customers, the cost of replicating the network from the point of customer connection to the high voltage network to the connection with the TransGrid transmission network was identified.
- Develop an estimate of the annual capital charge for the relevant network by applying the same return to capital and depreciation parameters as set out in the Determination.
- Develop an estimate of the operating costs that would be incurred in order to provide services to the tariff class by identifying those operating costs that are necessarily incurred by EnergyAustralia for the provision of network services to that tariff class.

The associated summation of the annual estimated capital costs and operating costs for each tariff class are therefore estimated to represent the standalone costs that would be incurred by a hypothetical new entrant.

The estimated standalone costs for each tariff class are set out in the following table.

Table 8: Comparison of Standalone Costs Vs FY11 DUOS Tariffs

Tariff Class	Total standalone costs FY11 (\$m)	Weighted average revenue FY11 (\$m)	Weighted average revenue as a proportion of standalone costs (%)
Domestic	\$1,405.24	\$744.6	53.0%
Low Voltage Business	\$1,384.45	\$672.4	48.6%
High Voltage	\$879.49	\$38.2	4.3%
Sub-trans.	\$598.13	\$5.3	<1%
CRNP	\$879.49	\$33.4	4%
Unmetered	\$1,290.89	\$9.9	<1%

It can be seen that costs need to establish and maintain a standalone service for each tariff class is significantly higher than the weighted average revenue to be recovered in FY11.

Estimating avoidable costs

The avoidable costs represent those costs that could be avoided by a business if it was not supplying goods or services to its customers. It represents the lower bound of costs that should be recovered from customers. If customers were charged below the avoidable costs it would be economically beneficial for the business to stop supplying the customer and thereby avoid the associated costs which would exceed the revenue expected to be obtained from the customer.

To estimate the avoidable costs for each proposed tariff class, EnergyAustralia has:

- identified those categories of costs that would be avoided if a tariff class was no longer served;
- allocated the cost of each avoidable cost category to each tariff class based on either volume or customer numbers, according to the nature of the cost category, to determine those costs that would be reduced if a particular tariff class was no longer served;
- summed the allocated avoidable cost for all avoidable cost categories for each tariff class to estimate the total avoidable cost for each tariff class.

The cost categories that were identified as being avoidable included:

- repairs and maintenance – this was identified as a partially avoidable cost given that use of the network impacts on its deterioration and the need for repairs and maintenance;
- customer service – these costs relate to managing billing and customer service enquiries which are related to the number of customers served;
- metering costs – these costs relate to meter reading and meter replacement and are related to the number of customers served;
- corporate and divisional support costs – costs relating to media, marketing and legal expenses can be considered to be driven by the number of customers served; and
- customer connections and installation inspections – these costs can be considered to be proportional to customers numbers, the voltage and capacity of the connection.

The following table sets out the avoidable costs for each of the tariff classes.

Table 9: Comparison of Avoidable Costs Vs FY11 DUOS Tariffs

Tariff Class	Total avoidable costs FY11 (\$m)	Weighted average distribution revenue FY11 (\$m)	Avoidable costs as a proportion of weighted average distribution revenue (%)
Domestic	\$177.97	\$744.6	23.9%
Low Voltage Business	\$57.55	\$672.4	8.6%
High Voltage	\$5.37	\$38.2	14.0%
Sub-trans.	\$3.34	\$5.3	62%
CRNP	\$10.99	\$33.4	33%
Unmetered	\$0.70	\$9.9	7%

It can be seen that in most cases the distribution revenue to be recovered for each tariff class is well above the avoidable cost for each tariff class. The sub-transmission tariff class has a lower weighted average revenue (per unit) when compared to the other tariff classes. This is because sub-transmission network tariffs are lower when compared to the other tariff classes. Hence the avoidable costs for sub-transmission services make up a higher proportion of the equivalent FY11 revenue.

As required under clause 6.18.2 of the NER, the following table provides a comparison of weighted average distribution revenue in FY11 with the estimates of standalone and avoidable costs for each tariff class. As required under the NER, the weighted average distribution revenue lies within the bounds of the upper limit (standalone cost) and the lower limit (avoidable cost).

Table 10: Efficient Pricing Bounds Test

Tariff Class	Total avoidable costs FY11 (\$m)	Weighted average distribution revenue FY11 (\$m)	Standalone costs FY11 (\$m)
Domestic	\$177.97	\$744.6	\$1,405.24
Low Voltage Business	\$57.55	\$672.4	\$1,384.45
High Voltage	\$5.37	\$38.2	\$879.49
Sub-trans.	\$3.34	\$5.3	\$598.13
CRNP	\$10.99	\$33.4	\$879.49
Unmetered	\$0.70	\$9.9	\$1,290.89

Recovery of Total Regulated Revenues

This section briefly describes the WAPC formula and the approach adopted to estimate consumption to be used in the formula when new tariffs are introduced, or customers move between tariffs.

The Weighted Average Price Cap (WAPC) formula governing EnergyAustralia's DUOS pricing is defined below:

$$\frac{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^t * q_{ij}^{t-2}}{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{t-1} * q_{ij}^{t-2}} \leq (1 + \Delta CPI) * (1 - X_t) * (1 + D_t)$$

$i=1,...,n$ and $j=1,...,m$

where:

- n is the number of distribution tariffs
- m is the number of distribution tariff components
- p_{ij}^t is the proposed price for component j of the distribution tariff i for Year t
- p_{ij}^{t-1} is the price charged by the DNSP for aggregate component j of the distribution tariff i in Year $t-1$ (ie the Year immediately preceding Year t)
- q_{ij}^{t-2} is the audited quantity of aggregate component j of the distribution tariff i charged by the DNSP in Year $t-2$ (ie the Year immediately prior to Year $t-1$)
- X_t is the value of “ X ” specified for the DNSP (for the Year t) by the AER
- ΔCPI is the December to December change in the Consumer Price Index from Year $t-2$ to Year $t-1$

An important feature of the WAPC formula is the use of lagged electricity consumption data to demonstrate compliance. Lagged data is used because of the delays in this data being audited and made available, and the uncertainty associated with the use of forecast volume data within the formula.

The use of lagged consumption data impacts on EnergyAustralia’s ability to recover its revenue requirement in circumstances when actual consumption in a year varies considerably from lagged consumption. This can occur when new tariffs or charging parameters are introduced, or when customers move between tariffs.

In these circumstances, EnergyAustralia applies a ‘reasonable estimates’ methodology to calculate the volume to be included in the WAPC formula. This ensures that revenue is maintained despite the tariff changes. For each year of the 2009 to 2014 distribution determination, EnergyAustralia will submit reasonable estimates of consumption to the AER for use in the WAPC formula if appropriate.

These estimates are based on the quantities that would have been sold in year $t-2$ if the new tariffs or tariff components had been introduced in that same year, or if EnergyAustralia mandated the transfer of customers from one tariff to another. Voluntary tariff transfers are not included in reasonable estimates.

Table 11: WAPC Notional Revenues (\$m)

Tariff Class	Weighted average distribution revenue FY10 (\$m)	Weighted average distribution revenue FY11 (\$m)	% Variation
Domestic	\$617.9	\$744.6	20.5%
LV Business	\$560.0	\$672.4	20.1%
HV Business	\$35.6	\$38.2	7.4%
Sub-transmission	\$4.7	\$5.3	12.7%
CRNP	\$30.1	\$33.4	11.0%
Unmetered	\$7.1	\$9.9	39.4%
Overall change in weighted average distribution revenue	\$1,255.4	\$1,503.9	19.8%

5. Variations to Tariffs

RULE REQUIREMENT

Clause 6.18.2(b)(5) of the NER requires that a pricing proposal set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur

The small number of customers affected by the decision to make these network tariffs redundant will be assigned to an appropriate network tariff from 1 July 2010 in accordance with EnergyAustralia's tariff assignment/re-assignment policy, as set out in Attachment G.

EnergyAustralia does not propose to vary or adjust the tariff during the course of the regulatory year.

In general, EnergyAustralia is not making any variations to the structure of network tariffs, such as adding or reforming of charging parameters. However, EnergyAustralia intends to make redundant the following network tariffs from 1 July 2010.

Table 12: Redundant Network Tariffs in FY11

Tariff Code	Tariff Description	Estimated Number of Customer
EA024	LV Residential < 40 MWh (No NAC)	0
EA026	LV Residential < 40 MWh (Substation)	0
EA224	LV Business < 40 MWh (No NAC)	0
EA226	LV Business < 40 MWh (Substation)	0
EA303	LV Business 40-160 MWh (Substation)	57
EA306	LV Business 160-750 MWh (Substation)	264
EA320	LV Business >750 MWh (Substation)	1,216

6: Climate Change Fund

RULE REQUIREMENT

Clause 6.18.2(b)(5A) of the NER requires that a pricing proposal must set out the amount paid to the Climate Change Fund in or in respect of the relevant regulatory year and reflect that amount in the expected revenue for the relevant regulatory year

For the year ending June 2011, EnergyAustralia will pay \$70,996,779 to the Climate Change Fund.

EnergyAustralia expects to recover \$70,996,779 for this purpose from network charges. Attachment F is the correspondence received from the Department of Environment and Climate Change on EnergyAustralia's contribution to the climate change fund for FY11. It is important to note that consultation is occurring within the NSW Government on EnergyAustralia's contribution to the Climate Change Fund in FY11.

7: Transmission Use of System Services

RULE REQUIREMENT

Clause 6.18.2(b)(6) of the NER requires that a pricing proposal must set out how charges for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year. In addition, clause 6.18.7 states that recovery of transmission revenue should not exceed the estimated amount of these charges for the relevant regulatory year, once the overs and unders account has been taken into account.

EnergyAustralia’s Transmission Use of System (TUOS) tariffs are designed to recover the allowed revenue for our transmission network, pass through the TransGrid transmission cost to customers and to recover (return) an under (over) recovery of transmission costs in the previous period. The process used by EnergyAustralia to achieve these TUOS pricing outcomes is summarised below:

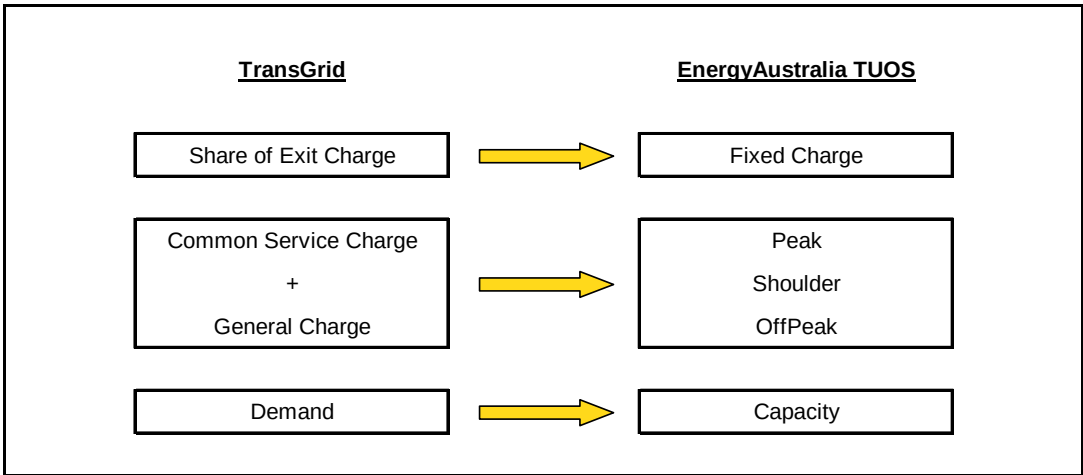
- Step 1: To calculate the annual amount of allowed revenue to recover for EnergyAustralia transmission assets;
- Step 2: EnergyAustralia to provide this annual revenue requirement to TransGrid for inclusion in their modelling of transmission charges for the coming year;

- Step 3: TransGrid to provide EnergyAustralia with the final transmission charges for the coming year in early April, as per the Memorandum of Understanding between the two organisations.
- Step 4: EnergyAustralia to set TUOS tariffs for the coming year to recover the forecast transmission cost plus the opening balance (positive or negative) of the transmission overs and unders account.

EnergyAustralia’s approach to setting Transmission Use Of System (TUOS) prices is based on the fundamental principle of preserving the TransGrid transmission price signal, where it is desirable to do so from an economic and equity perspective. Given the location-basis of TransGrid’s transmission charges, it is only possible to preserve the transmission price signal in the TUOS component of network tariffs for Cost Reflective Network Price (CRNP) customers given the site-specific nature of these network tariffs.

As illustrated in the following figure, EnergyAustralia’s approach to setting TUOS tariffs for CRNP customers preserves the TransGrid transmission price signal, in spite of the structural differences between the two price signals.

Figure 5: EnergyAustralia’s TUOS Pricing Methodology for CRNP Customers



While it is not possible to provide individual customers on a published network tariff with cost reflective TUOS prices due to “postage stamp” basis of these tariffs, it is possible, however, to allocate annual TUOS cost (net of CRNP TUOS recovery) to each tariff class on an equitable basis using cost allocation principles. As with any cost allocation process, this approach requires a significant degree of averaging of costs and assumptions to be

made in relation to the time of use consumption and demand characteristics of tariff classes where interval consumption data does not exist. The following table provides a comparison of the forecast TUOS revenue expected to be recovered from each tariff class in FY11 with the forecast transmission cost allocated to each tariff class in FY11.

Table 13: Comparison of TUOS Revenue Vs Cost Allocation – FY11

Tariff Class	Forecast TUOS Revenue (\$m)	Forecast Transmission Cost Allocation (\$m)	Forecast Variation (%)
Residential	\$133.6	\$155.9	-14.3%
Low Voltage Business	\$244.2	\$211.9	15.2%
High Voltage	\$22.1	\$23.3	5.2%
Sub-transmission Voltage	\$12.5	\$13.3	-6.0%
CRNP	\$43.3	\$50.5	-14.3%
Unmetered	\$2.7	\$3.5	-23.9%
Total	\$458.4	\$458.4	-

It is clear from the above table that there are differences between the forecast TUOS revenue and the allocated transmission cost at the individual tariff class level in FY11. These differences arise because of the following reasons:

- it is not possible to preserve the TransGrid transmission price signal for customers on published network tariffs;
- The likelihood of forecast volume error;
- The NER requirement to set TUOS tariffs to achieve a zero forecast balance of the transmission overs and unders account by the end of the year that the new prices apply; and
- EnergyAustralia's commitment to transitioning TUOS prices to cost reflective levels, where necessary to avoid imposing unacceptable price shocks on individual customers, as discussed below.

A major development in transmission pricing is TransGrid decision to remove the peak and shoulder energy charges from its transmission charges in FY11. This significant rebalancing of transmission charges resulted in the cost reflective TUOS prices increasing by well above the average X factor increase for some individual CRNP customers. While it is important from an economic perspective to preserve the TransGrid price

signal to these customers, EnergyAustralia believes that it is unfair to burden any individual customer with unacceptable price shocks arising from unexpected changes in the structure of prices. It is for this reason that EnergyAustralia has decided to transition TUOS tariffs to cost reflective levels over a reasonable period for a small number of CRNP customers. It is important to note that the TUOS transitional pricing arrangements for the small number of CRNP customers is expected result in an immaterial distortion to TUOS prices for published tariff customers over the next two years. Please refer to Attachment L for a conceptual discussion on EnergyAustralia's TUOS transitional arrangements.

Unders and overs account

The over and unders transmission account is a mechanism to ensure that if transmission revenue recovery is too high or too low in a given regulatory year, Distribution Network Service Providers, such as EnergyAustralia can recover the difference between actual transmission revenue and transmission payments by adjusting TUOS prices in the subsequent year. Fluctuations in revenue recovery are most likely to be caused by the impact of weather on electricity consumption and demand forecasts.

As required under the NER, EnergyAustralia have set TUOS tariffs to achieve a forecast zero balance of the overs and unders account for transmission by the end of FY11, as shown in the following table:

Table 14: Overs and Unders Account Forecast Closing Balance

Financial year ending	t-2 audited FY09	t-1 expected FY10	t forecast FY11
Interest rate applicable to balance %	9.70%	10.02%	10.02%
Opening balance \$ 000	(5,989)	(12,210)	(619)
Interest on opening balance (365 days) %	581	(1,223)	(62)
Forecast over / (under) recovery for financial year	(5,380)	(12,217)	650
Interest charged on over/under recovery for financial year \$000	(261)	(597)	32
Closing balance of transmission overs/ (unders) account \$000	(12,210)	(619)	0

8: Changes from the Previous Regulatory Year

RULE REQUIREMENT

Clause 6.18.2(b)(8) of the NER requires that a pricing proposal must describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the Rules and any applicable distribution determination

This chapter sets out the principal changes since the previous regulatory year, and demonstrates that these changes comply with the Rules and the AER's distribution and transmission determinations released on 30 April 2009.

EnergyAustralia has already noted in Chapter 5 that it intends to make redundant certain network tariffs from 1 July 2010. The small number of customers affected by the decision to make these network tariffs redundant will be assigned to an appropriate network tariff from 1 July 2010 in accordance with EnergyAustralia's tariff assignment/re-assignment policy, as set out in Attachment G.

Compliance of tariffs and charging parameters with the Determination

On the 30 April 2009 the AER made its final determination approving the largest capital and

operating investment programs in EnergyAustralia's history. These programs are required to renew the network and ensure continued energy supply to our customers.

Subsequently, EnergyAustralia and the other two NSW distribution network businesses (Integral Energy and Country Energy) sought a review of the AER's decision. On 25 November 2009, the Australian Competition Tribunal varied the AER Determination by increasing the nominal vanilla WACC from 8.80% to 10.02%, increased controllable operating expenditure allowance for EnergyAustralia by \$4.5m, amending the definition of the general nominated pass-through event for EnergyAustralia and remitting the AER's decision in respect of EnergyAustralia's alternative control (public lighting) services with a re-determination by the AER by 15 April 2010.

As a result of this outcome, the new AER approved real increase (ie X Factors) over the remainder of the current Regulatory Period to apply to EnergyAustralia's distribution prices and transmission revenue have been varied from those set out in the AER Determination released on 30 April 2009. The new X-factors are the principal change since the previous regulatory year and are reproduced in the following table.

Table 15: EnergyAustralia's Annual X Factor – Remainder of Regulatory Period

	FY10	FY11	FY12	FY13	FY14
Forecast CPI	2.47%	2.47%	2.47%	2.47%	2.47%
Distribution	-17.86%	-18.18%	-18.18%	-18.18%	0.77
Transmission	-7.77%	-18.46%	-18.46%	-18.46%	-2.02

Compliance of tariffs and charging parameters with the Rules

Rule 6.18.6 provides that the weighted average revenue per tariff class for standard control services cannot change by more than two percent from the preceding year, after adjustments for the X-factor and the consumer price index have been applied.

To demonstrate compliance with the Determination and the Rules, EnergyAustralia has provided the AER with a compliance spreadsheet that sets out the detailed assumptions used to apply the weighted average price cap (WAPC). These assumptions include:

- Audited usage volumes and customer numbers for the year ending June 2009
- adjustments to usage volumes and customer numbers for reasonable estimates

- Distribution use of system (DUOS), transmission use of system (TUOS) and Climate Change Fund rates by tariff for the year ending June 2011
- network use of system (NUOS) rates by tariff for the year ending June 2011
- revenues by tariff and tariff class for the years ending June 2010 and 2011

The compliance spreadsheet included as Attachment D demonstrates EnergyAustralia's compliance with the Determination and the Rules.

EnergyAustralia has also commissioned the National Economic Research Associates(NERA) to review this document and the compliance spreadsheet from a compliance perspective. Refer to Attachment H for NERA's correspondence.

As required, EnergyAustralia engaged PriceWaterhouseCoopers to undertake an independent review of the WAPC variables included in the Schedules. The statement from PriceWaterhouse Coopers is provided in Attachment I.

9. Public Lighting Services

EnergyAustralia's regulated services are known as "direct control services" meaning that they are subject to a distribution determination. Direct control services are further classified into "standard" control services and "alternative control services". Public Lighting Services are classified as alternative control services and are subject to a different control mechanism to general network services which are classified as standard control services.

The AER made its final decision for EnergyAustralia's public lighting services on 15 April 2010. That decision applies a control mechanism from which the prices or tariffs to be applied to each of EnergyAustralia's public lighting customer are derived. Whilst Clause 6.18 of the Transitional Chapter 6 Rules, applies to EnergyAustralia's public lighting services, the application is limited given that the tariffs can be derived directly from the AER's decision. This chapter explains the application of these pricing rules to EnergyAustralia's public lighting services in the context of the decision which has been made by the AER.

Tariffs and tariff classes

All public lighting customers are subject to the tariffs in the AER April 2010 determination and therefore are defined to be a single tariff class.¹⁰ The tariffs that apply to these customers are:

- A fixed capital charge for installed public lighting assets. The fixed capital charges have been provided in confidential Attachment C because they are individual customer charges. This tariff only relates to the capital charges for public lighting assets constructed before 1 July 2009.

- An annuity price per asset type for all new assets. These tariffs and charging parameters are contained in Attachment B. This price relates to the capital charge for new public lighting assets constructed after 30 June 2009, but does not apply to assets where the customer has funded the original construction cost.
- A maintenance price per asset. This tariff relates to the maintenance of each of EnergyAustralia's public lights. These tariffs and charging parameters are contained in Attachment B.

Adjustments

The AER April 2010 public lighting decision and clause 6.18.2(b)(5) of the transitional Rules require EnergyAustralia to explain any adjustment to prices that could occur during the regulatory year. EnergyAustralia has made one adjustment to the prices that will be levied from 1 July 2010 to 30 June 2011.

EnergyAustralia's January 2010 proposal mixed up the names of three assets in the model that determined the annuity price for assets constructed after 30 June 2009. This carried through into the AER's April 2010 determination and inadvertently approves a 592% increase in the annuity price of the "50W MBF – PLAIN" luminaire. The following table shows the approved prices and EnergyAustralia's adjusted prices for these three assets.

EnergyAustralia has also adjusted the FY11 fixed capital charges as some customers paid residual value (\$0.2m) for assets replaced early during 2010. As a result we have deducted this amount from the 2010 closing RAB value. The adjustment is a minor reduction in the fixed capital charge for the relevant customers.

¹⁰ Tariff class: A class of customers for one or more *direct control services* who are subject to a particular tariff or particular tariffs.

Table 16: 50 watt mercury vapour annuity capital charges

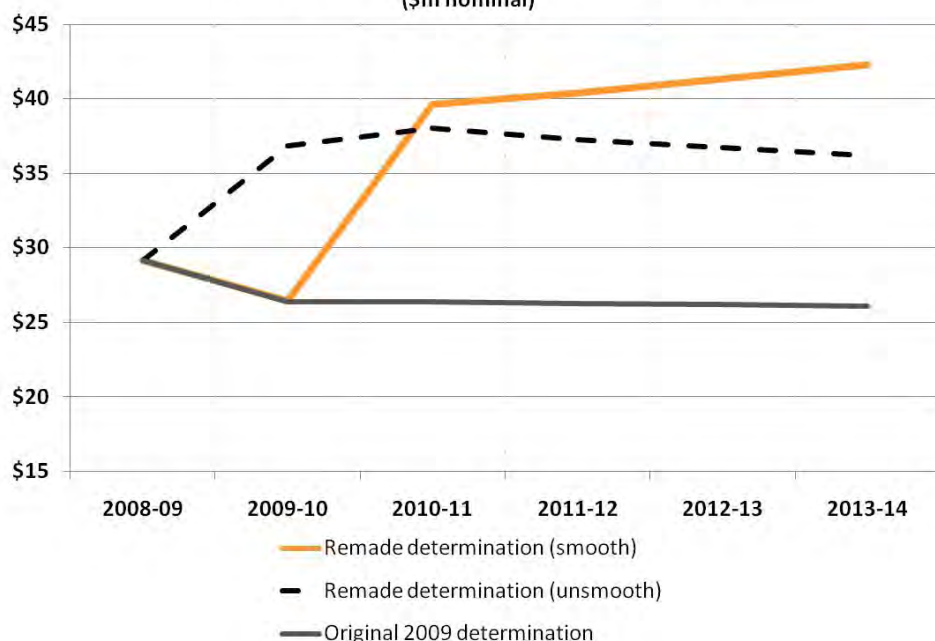
Luminaire	FY10 actual charges	FY11 approved charges	Approved increase	FY11 amended prices	Amended increase
50W MBF - BOURKE HILL	\$69.97	\$13.13	-81%	\$80.62	15%
50W MBF - NOSTALGIA	\$68.52	\$80.62	18%	\$78.94	15%
50W MBF - PLAIN	\$11.40	\$78.94	592%	\$13.13	15%

Impact on revenue

During FY10 EnergyAustralia will recover approximately \$26m in public lighting charges, based on the control mechanism in the AER's April 2009 determination. This was a 10% reduction in total charges from the previous year's charges of \$29m. The AER has now remade its

determination and on the basis of its new determination EnergyAustralia was entitled to recover approximately \$37m in FY10. That is, an additional \$11m should have been charged during 2009-10 that was not reflected in customer bills. Had this \$11m been included in customer bills, there would have been a 26% increase in charges from FY09

Figure 6: Actual and allowed public lighting revenue (\$m nominal)



The remade determination allows the under-recovered \$11m to be recovered over FY11 to FY14. Therefore, the 10% reduction in 2009-10 charges was an artificial reduction based on input assumptions that have now been corrected in the AER's April 2010 determination. This artificial price reduction now needs to be offset by a 50% increase in FY11 charges. This will allow EnergyAustralia to more closely align its revenue from public lighting services with the costs of providing those services.

The majority of the increase is necessary as the charges recovered over the past regulatory period were constrained. The constraint was that only a small proportion the capital invested was recovered through prices, which meant EnergyAustralia has carried this capital investment forward in its regulatory asset base value and is the basis for the increased customer charges.

Impact on maintenance prices

The remade determination will have a varied impact on prices. In relation to maintenance prices, the opex modelling for the FY10 prices was an allocation of a target operating expenditure and the price of maintaining street lights was highly averaged. The remade determination was based on a build up of costs, which aimed to develop more cost reflective maintenance prices. This change in approach from FY10 to FY11 has resulted in prices moving significantly, with some increases and some decreases in asset maintenance prices. However, it must also be recognised that there is a general upward price movement of 4%.¹¹

This general increase is mainly due to artificial prices being charged in FY10. The maintenance prices levied in FY10 were based on achieving a target of \$13.2m¹² for maintenance. In reality, the maintenance prices delivered 4% less revenue than the target and only contributed \$12.7m. This issue was corrected by the AER when it remade its determination and the new maintenance prices are expected to recover \$13.9m in public lighting revenue.

Impact on annuity prices

On average the annuity price associated with the construction of new public lights has increased by 19% since FY10.¹³ This is due to the use of a WACC of 8.1% rather than 6.8% (real pre-tax) in the AER's April 2010 determination. This large price increase is also due to an artificial price reduction allowed in the FY10 year, where prices for the construction of new assets fell by 29%.¹⁴

Other requirements of determination

The transitional Rules require the pricing proposal to demonstrate compliance with the distribution determination. The AER's remade determination for public lighting states that EnergyAustralia should provide actual public lighting charges for the current year in the pricing proposal.¹⁵ The following two tables show the total charges recovered in FY10 and the expected charges in FY11. It should be noted that the actual the actual charges for FY10 have been prorated from the charges from July 2009 to March 2010 and the expected 2010 charges have been calculated against the public lighting inventory as at March 2010.

¹¹ This is the average price change per component weighted by the number of components.

¹² AER, *AER revised - EA Annuity Model T3 T4.xls*, 1 April 2009, sheet "Inputs – General", cell "C49"

¹³ This is the average price change per component weighted by the number of components.

¹⁴ This is the average price change per component weighted by the number of components.

¹⁵ AER, *EnergyAustralia distribution determination 2009-10 to 2013-14, Alternative control (public lighting) services*, 13 April 2010, page 54.

Table 17: FY10 customer charges (estimate to year end)

Charges for pre-July 09 assets	Charges for post-July 09 assets	Residual value charges for early replacement of pre-July 09 assets	Total FY10 customer charges
\$24.6m	\$1.8m	\$0.2m	\$28.4m

Table 18: FY11 customer charges (forecast \$ m nominal)

Maintenance for assets constructed by March 2010	Capital charges for post-June 09 assets	Total
\$13.9m	\$25.7m	\$39.6m

Other rule requirements

The transitional Rules require EnergyAustralia to demonstrate that each tariff class' expected revenue be between stand alone cost and avoidable of providing the service. The Rules also require EnergyAustralia to take into account the long run marginal cost for each charging parameter, and have regard for transaction costs and whether relevant tariff class is able to respond to price signals. These Rule requirements so far as they applicable to public lighting services should be considered as addressed through the implementation of the AER's determination, which was based on the AER's view of efficient cost.

The other transitional Rule requirement that is applicable to direct control services is the assignment of customers to tariff classes (6.18.4). As public lighting is its own tariff class and there have been no-assignment of customers, this Rule obligation has been met for public lighting services for this year. All public lighting customers are subject to the same three public lighting tariffs, EnergyAustralia is satisfied that the criteria in transitional Rule 6.18.4 have been properly accounted for.

10. Proposed New Network Tariffs

EnergyAustralia is planning to undertake a pricing trial program during the course of FY11 either as a result of a successful bid to the Commonwealth Government's Smart Grid Smart City initiative, or as part of EnergyAustralia's on-going customer research program. It is important to note that these trials will be on based on the voluntary participation of customers with participation in the trial capped at around 50,000 customers.

EnergyAustralia's preferred approach is to offer participating retailers innovative network tariffs, which are then passed through to end-customers. The alternative approach is to offer participating retailers standard network tariffs and enter into costly and complex contractual arrangements to require retailers adjust the retail-component of bundled electricity prices to convey the desired price signals to end-customers.

Given the current uncertainty surrounding the timing and scope of these pricing trials, EnergyAustralia is not able to submit final network pricing trial tariffs to the AER to consider as part of this pricing proposal. Nevertheless, EnergyAustralia has a clear idea of the number and structure of the network tariffs that it wishes to test, and which tariff class applies to these proposed new network tariffs. Therefore, EnergyAustralia wishes to adjust these proposed new network tariffs during the course of FY11 once the arrangements with retailers and the trial design have been finalised. To use the proposed trial network tariffs set out in this trial without adjustment could undermine the objectives of the pricing trial. Therefore EnergyAustralia seeks the support of the AER in undertaking these pricing trials given the importance of providing the industry with a better understanding of the acceptance and response of customers and retailers to innovative pricing solutions.

Description of Proposed new network tariffs

The table below provides a summary of the description of the proposed network tariffs for the purposes of undertaking a voluntary pricing trial program. Given that these tariffs are intended to be used for research purposes, together with the potential for these tariffs to be adjusted during the course of FY11, EnergyAustralia believes that it is not appropriate to publish these tariffs.

Proposed Adjustment to these tariff

To ensure that the proposed new network tariffs do not undermine the achievement of the objectives of the pricing trial, EnergyAustralia wishes to have the option of make a variation or adjustment to these tariffs within the course of FY 2011. The proposed change is set out below:

- To lower the level of the DUOS price of the proposed new network tariffs to ensure adequate voluntary take-up of the pricing trial is achieved.

It is important to note that EnergyAustralia does not change the tariff structure or the TUOS prices of the proposed new network tariffs during the course of FY11.

Compliance with the NER

The TUOS component of the proposed new network tariffs included in the pricing proposal for FY11 are based on proposed TUOS component for EnergyAustralia's residential Time of Use tariff (EA25) for FY11. The DUOS component of the proposed new network tariffs have been developed with the expectation of a downward adjustment during the course of FY11, as discussed above. EnergyAustralia intends to provide the AER with any change within FY11 to the DUOS component of these network tariffs, for consideration from a compliance perspective, prior to the implementation

Table 19: Description of the Proposed New Network Tariffs for FY11 – Pricing Trial Program

	Measure	Objective	What is the customer being asked to do?	What enablers will they have?	How will they be rewarded?
1	Dynamic Peak Pricing	Reduce demand during critical peak periods to avoid or defer capital expenditure. Reduce demand during periods of abnormally high prices for supplied electricity.	Select and turn off appliances during a peak period in response to notification from the utility.	Call informing them of the peak event. Feedback on real-time energy use. Information/advice on what to switch off.	Reduced energy bill by lowering energy consumption during a high-priced period.
2	Dynamic Peak Rebate	Reduce demand during critical peak periods to avoid or defer capital expenditure. Reduce demand during periods of abnormally high prices for supplied electricity.	Select and turn off appliances during a peak period in response to notification from the utility.	Call informing them of the peak event. Feedback on real-time energy use. Information/advice on what to switch off.	Receive a rebate on their energy bill by lowering energy consumption during a critical peak period.
3	Interruptible Load Tariff	Reduce demand during critical peak periods to avoid or defer capital expenditure. Reduce demand during periods of abnormally high prices for supplied electricity.	Allow utility to turn off selected household appliances during a critical peak period. Once off decision to subscribe.	Automatic appliance control (no effort required) Feedback on real-time energy use	Reduced energy bills due to lower tariff charged by utility.
4	Interruptible Load Rebate or Reward	Reduce demand during critical peak periods to avoid or defer capital expenditure. Reduce demand during periods of abnormally high prices for supplied electricity.	Allow utility to turn off selected household appliances during a critical peak period. Once off decision to subscribe. Automatic appliance control (no effort required).	Feedback on real-time energy use.	Reward may be financial (direct deposit, cheque, rebate on next energy bill) or non-financial such as movie tickets.
5	Air Conditioning Cycling	Reduce demand during peak periods to avoid or defer capital expenditure. Reduce the average price of supplied electricity.	Allow utility to cycle air conditioners for a set period each day. Once off decision to subscribe. Automatic appliance control (no effort required)	Feedback on real-time energy use of home appliances.	Receive a rebate on their energy bill for providing this control to the utility during peak events.
6	Seasonal Time-of-Use Tariff	Reduce demand during peak periods to avoid or defer capital expenditure. Reduce the average price of supplied electricity.	Select and turn off appliances during peak periods.	Feedback on real-time energy use. Information/advice on what to switch off.	Reduced energy bills depending on level of response.
7	Discretionary price response linked to advanced IHD	Reduce demand during periods of abnormally high prices for supplied electricity.	Programme an advanced IHD to automatically adjust energy usage in response to price signals	Advanced IHD and call informing them of the pricing event.	Reduced energy bills depending on level of response.
8	Pre-payment Tariff	Lower credit risk and increase cash flow. Reduced greenhouse gas emissions.	Pay for electricity use in advance and keep usage below credit level. Identify ways to limit household electricity use throughout the year.	Feedback on energy use and remaining credit. Information/advice on ways to save.	Easier budget management. Reduced energy bills and greenhouse gas emissions by limiting their energy consumption.

11. Attachments

Attachments have been included with this pricing proposal as follows:

Attachment	Disclosure	Description
A	CONFIDENTIAL	Price and Tariff Class Information for Cost Reflective Network Price (CRNP) Customers
B	CONFIDENTIAL	Customer Bill Information for Public Lighting Customers
C		Price Information for Public Lighting Tariffs
D	CONFIDENTIAL	Completed Compliance Spreadsheet
E		Price and Tariff Class Information for Miscellaneous & Monopoly Charges and Emergency Recoverable Works
F		Notification of Climate Change Fund Contribution
G		Procedure on Assignment and Re-assignment of customers to tariffs
H	CONFIDENTIAL	Statement from NERA Economic Consulting
I	CONFIDENTIAL	Statement from PriceWaterhouseCoopers Australia
J		TransGrid's Confirmation of its Revenue Recovery for FY11
K	CONFIDENTIAL	Economic Cost Model
L	CONFIDENTIAL	Proposed TUOS Transitional Arrangements

Attachment A: Cost Reflective Network Prices (Confidential)

Attachment B: Public Lighting Schedule of Charges for Existing Components (Confidential)

Attachment C: Public Lighting Prices for New Components

Asset	Prices to be levied in FY11		
	Maintenance	Capital	Total
Bracket - 0.5	\$0.00	\$17.40	\$17.40
Bracket - 0.6	\$0.00	\$17.40	\$17.40
Bracket - 1.0	\$0.00	\$16.61	\$16.61
Bracket - 1.2	\$0.00	\$16.61	\$16.61
Bracket - 1.5	\$0.00	\$66.27	\$66.27
Bracket - 2.0	\$0.00	\$21.58	\$21.58
Bracket - 2.5	\$0.00	\$26.80	\$26.80
Bracket - 3.0	\$0.00	\$41.70	\$41.70
Bracket - 3.5	\$0.00	\$43.95	\$43.95
Bracket - 4.0	\$0.00	\$43.95	\$43.95
Bracket - 4.5	\$0.00	\$50.27	\$50.27
Bracket - 5.0	\$0.00	\$48.24	\$48.24
Bracket - 6.0	\$0.00	\$66.30	\$66.30
Bracket - 6.5	\$0.00	\$66.30	\$66.30
Bracket - 7.0	\$0.00	\$66.30	\$66.30
Bracket - 8.0	\$0.00	\$66.30	\$66.30
Bracket - C4	\$0.00	\$87.52	\$87.52
Bracket - NIL	\$0.00	\$0.00	\$0.00
Bracket - PRIVATE	\$0.00	\$10.52	\$10.52
Bracket - SUSPENDED	\$0.00	\$58.31	\$58.31
Bracket - T1	\$0.00	\$28.86	\$28.86
Bracket - T2	\$0.00	\$47.57	\$47.57
Bracket - T2A	\$0.00	\$47.57	\$47.57
Bracket - T3	\$0.00	\$48.24	\$48.24
Bracket - T3A	\$0.00	\$48.24	\$48.24
Bracket - T4	\$0.00	\$46.21	\$46.21
Bracket - T5	\$0.00	\$46.21	\$46.21
Bracket - T6	\$0.00	\$66.30	\$66.30
Bracket - T7	\$0.00	\$60.54	\$60.54
Connection - OH	\$0.00	\$0.00	\$0.00
Connection - OH2	\$0.00	\$0.00	\$0.00
Connection - OHS	\$0.00	\$0.00	\$0.00
Connection - O/U	\$81.66	\$10.60	\$92.26
Connection - UG2	\$0.00	\$0.00	\$0.00
Connection - UGORDA	\$40.83	\$10.60	\$51.43
Connection - UGR1	\$74.85	\$14.70	\$89.55
Connection - UGR2	\$27.22	\$10.60	\$37.82
Connection - UGS	\$0.00	\$10.60	\$10.60
Connection - UG-SP	\$0.00	\$0.00	\$0.00
Lamp - EMPTY	\$35.47	\$0.00	\$35.47
Lamp - INC1x100	\$194.85	\$0.00	\$194.85
Lamp - INC1x1000	\$371.84	\$0.00	\$371.84
Lamp - INC1x1440	\$193.47	\$0.00	\$193.47
Lamp - INC1x150	\$199.02	\$0.00	\$199.02
Lamp - INC1x200	\$200.80	\$0.00	\$200.80
Lamp - INC1x300	\$221.21	\$0.00	\$221.21
Lamp - INC1x40	\$194.94	\$0.00	\$194.94
Lamp - INC1x500	\$252.91	\$0.00	\$252.91
Lamp - INC1x60	\$194.85	\$0.00	\$194.85
Lamp - INC1x75	\$194.85	\$0.00	\$194.85
Lamp - INC3x100	\$196.49	\$0.00	\$196.49
Lamp - MBF1x1000	\$97.60	\$0.00	\$97.60
Lamp - MBF1x125	\$45.98	\$0.00	\$45.98
Lamp - MBF1x160	\$39.79	\$0.00	\$39.79
Lamp - MBF1x250	\$46.12	\$0.00	\$46.12
Lamp - MBF1x400	\$46.01	\$0.00	\$46.01
Lamp - MBF1x42	\$44.28	\$0.00	\$44.28
Lamp - MBF1x50	\$38.23	\$0.00	\$38.23
Lamp - MBF1x500	\$115.99	\$0.00	\$115.99
Lamp - MBF1x700	\$70.19	\$0.00	\$70.19
Lamp - MBF1x80	\$40.30	\$0.00	\$40.30
Lamp - MBF1x800	\$115.99	\$0.00	\$115.99
Lamp - MBF2x125	\$55.18	\$0.00	\$55.18
Lamp - MBF2x160	\$52.26	\$0.00	\$52.26
Lamp - MBF2x175	\$104.16	\$0.00	\$104.16

Asset	Prices to be levied in FY11		
	Maintenance	Capital	Total
Lamp - MBF2x400	\$53.25	\$0.00	\$53.25
Lamp - MBF2x80	\$45.13	\$0.00	\$45.13
Lamp - MBF3x160	\$52.26	\$0.00	\$52.26
Lamp - MBF3x250	\$60.82	\$0.00	\$60.82
Lamp - MBF3x400	\$60.49	\$0.00	\$60.49
Lamp - MBF3x80	\$49.97	\$0.00	\$49.97
Lamp - MBF4x1000	\$196.41	\$0.00	\$196.41
Lamp - MBF4x80	\$54.80	\$0.00	\$54.80
Lamp - MBF6x125	\$93.99	\$0.00	\$93.99
Lamp - MBF6x160	\$52.26	\$0.00	\$52.26
Lamp - MBF9x160	\$52.26	\$0.00	\$52.26
Lamp - MBI1x750	\$0.00	\$0.00	\$0.00
Lamp - MBI1x100	\$68.59	\$0.00	\$68.59
Lamp - MBI1x1000	\$159.21	\$0.00	\$159.21
Lamp - MBI1x150	\$98.49	\$0.00	\$98.49
Lamp - MBI1x1500	\$131.70	\$0.00	\$131.70
Lamp - MBI1x250	\$61.29	\$0.00	\$61.29
Lamp - MBI1x3745	\$74.77	\$0.00	\$74.77
Lamp - MBI1x400	\$59.66	\$0.00	\$59.66
Lamp - MBI1x500	\$116.19	\$0.00	\$116.19
Lamp - MBI1x70	\$72.00	\$0.00	\$72.00
Lamp - MBI2x400	\$80.56	\$0.00	\$80.56
Lamp - MBI4x150	\$277.65	\$0.00	\$277.65
Lamp - SON1x100	\$53.36	\$0.00	\$53.36
Lamp - SON1x1000	\$88.05	\$0.00	\$88.05
Lamp - SON1x120	\$71.17	\$0.00	\$71.17
Lamp - SON1x150	\$52.04	\$0.00	\$52.04
Lamp - SON1x220	\$83.01	\$0.00	\$83.01
Lamp - SON1x250	\$52.31	\$0.00	\$52.31
Lamp - SON1x310	\$82.03	\$0.00	\$82.03
Lamp - SON1x360	\$66.01	\$0.00	\$66.01
Lamp - SON1x400	\$54.28	\$0.00	\$54.28
Lamp - SON1x50	\$45.21	\$0.00	\$45.21
Lamp - SON1x70	\$46.50	\$0.00	\$46.50
Lamp - SON2x250	\$65.62	\$0.00	\$65.62
Lamp - SON2x400	\$69.54	\$0.00	\$69.54
Lamp - SON2x70	\$57.53	\$0.00	\$57.53
Lamp - SON3x70	\$68.55	\$0.00	\$68.55
Lamp - SON4x250	\$92.22	\$0.00	\$92.22
Lamp - SON4x600	\$133.54	\$0.00	\$133.54
Lamp - SON4x70	\$79.58	\$0.00	\$79.58
Lamp - SON8x70	\$123.69	\$0.00	\$123.69
Lamp - SOX1x135	\$51.85	\$0.00	\$51.85
Lamp - SOX1x150	\$86.31	\$0.00	\$86.31
Lamp - SOX1x180	\$170.33	\$0.00	\$170.33
Lamp - SOX1x90	\$74.96	\$0.00	\$74.96
Lamp - TF1x16	\$89.48	\$0.00	\$89.48
Lamp - TF1x176	\$119.77	\$0.00	\$119.77
Lamp - TF1x20	\$90.29	\$0.00	\$90.29
Lamp - TF1x236	\$119.77	\$0.00	\$119.77
Lamp - TF1x26	\$90.37	\$0.00	\$90.37
Lamp - TF1x40	\$90.44	\$0.00	\$90.44
Lamp - TF1x60	\$91.21	\$0.00	\$91.21
Lamp - TF1x80	\$91.21	\$0.00	\$91.21
Lamp - TF2x14 T5	\$42.21	\$0.00	\$42.21
Lamp - TF2x20	\$52.88	\$0.00	\$52.88
Lamp - TF2x26	\$90.79	\$0.00	\$90.79
Lamp - TF2x40	\$145.41	\$0.00	\$145.41
Lamp - TF2x58	\$89.48	\$0.00	\$89.48
Lamp - TF2x80	\$146.95	\$0.00	\$146.95
Lamp - TF3x20	\$91.07	\$0.00	\$91.07
Lamp - TF3x40	\$200.37	\$0.00	\$200.37
Lamp - TF3x80	\$202.69	\$0.00	\$202.69
Lamp - TF4x20	\$91.46	\$0.00	\$91.46
Lamp - TF4x40	\$255.34	\$0.00	\$255.34
Lamp - TF4x80	\$258.43	\$0.00	\$258.43
Lamp - TF5x58	\$89.48	\$0.00	\$89.48
Lamp - TF5x65	\$89.48	\$0.00	\$89.48
Lamp - TF5x80	\$314.17	\$0.00	\$314.17

Asset	Prices to be levied in FY11		
	Maintenance	Capital	Total
Lamp - TF6x20	\$92.24	\$0.00	\$92.24
Lamp - TF6x36	\$92.75	\$0.00	\$92.75
Lamp - TF6x80	\$369.91	\$0.00	\$369.91
Lamp - TH1x1000	\$75.21	\$0.00	\$75.21
Lamp - TH1x1500	\$72.93	\$0.00	\$72.93
Lamp - TH1x400	\$81.31	\$0.00	\$81.31
Lamp - TH1x500	\$68.78	\$0.00	\$68.78
Lamp - TH1x750	\$76.01	\$0.00	\$76.01
Luminaire - 1000W MBF	\$0.00	\$34.33	\$34.33
Luminaire - 1000W SON	\$0.00	\$183.17	\$183.17
Luminaire - 1000W SON FLOODLIGHT	\$0.00	\$95.75	\$95.75
Luminaire - 1000W/1500W MBI FLOODLIGHT	\$0.00	\$139.06	\$139.06
Luminaire - 100W MBI	\$0.00	\$29.66	\$29.66
Luminaire - 100W MBI FLOODLIGHT	\$0.00	\$34.33	\$34.33
Luminaire - 100W SON	\$0.00	\$25.57	\$25.57
Luminaire - 100W SON - PARKVILLE	\$0.00	\$136.23	\$136.23
Luminaire - 100W SON FLOODLIGHT	\$0.00	\$61.36	\$61.36
Luminaire - 100W SON -PLAIN	\$0.00	\$25.57	\$25.57
Luminaire - 125W MBF	\$0.00	\$13.42	\$13.42
Luminaire - 125W MBF - BOURKE HILL	\$0.00	\$93.21	\$93.21
Luminaire - 125W MBF - HYDE PARK	\$0.00	\$65.75	\$65.75
Luminaire - 125W MBF - NOSTALGIA	\$0.00	\$95.52	\$95.52
Luminaire - 125W MBF - PARKVILLE	\$0.00	\$122.22	\$122.22
Luminaire - 125W MBF BOLLARD	\$0.00	\$55.47	\$55.47
Luminaire - 125W MBF -PLAIN	\$0.00	\$13.42	\$13.42
Luminaire - 125W/250W MBF FLOODLIGHT	\$0.00	\$31.21	\$31.21
Luminaire - 135W SOX	\$0.00	\$37.45	\$37.45
Luminaire - 150W SON	\$0.00	\$24.75	\$24.75
Luminaire - 150W SON - HYDE PARK	\$0.00	\$65.75	\$65.75
Luminaire - 150W SON - PARKVILLE	\$0.00	\$136.23	\$136.23
Luminaire - 150W SON - PARKWAY 1	\$0.00	\$45.46	\$45.46
Luminaire - 150W SON FLOODLIGHT	\$0.00	\$2.07	\$2.07
Luminaire - 150W SON GEC 'BOSTON 3'	\$0.00	\$122.22	\$122.22
Luminaire - 150W/250W MBI FLOODLIGHT	\$0.00	\$81.88	\$81.88
Luminaire - 180W SOX	\$0.00	\$44.52	\$44.52
Luminaire - 1x40W TF	\$0.00	\$12.68	\$12.68
Luminaire - 1x80W TF	\$0.00	\$10.15	\$10.15
Luminaire - 250W MBF	\$0.00	\$23.70	\$23.70
Luminaire - 250W MBF - PARKVILLE	\$0.00	\$126.04	\$126.04
Luminaire - 250W MBF - PARKWAY 1	\$0.00	\$45.46	\$45.46
Luminaire - 250W MBI - SMARTPOLE	\$0.00	\$2.07	\$2.07
Luminaire - 250W SON	\$0.00	\$23.19	\$23.19
Luminaire - 250W SON - PARKVILLE	\$0.00	\$149.25	\$149.25
Luminaire - 250W SON - PARKWAY 1	\$0.00	\$45.46	\$45.46
Luminaire - 250W SON FLOODLIGHT	\$0.00	\$54.61	\$54.61
Luminaire - 250W SON GEC 'BOSTON 3'	\$0.00	\$125.05	\$125.05
Luminaire - 2X14W TF - T5 PIERLIGHT	\$0.00	\$19.53	\$19.53
Luminaire - 2x14W TF - T5 PIERLITE MK 3	\$0.00	\$29.35	\$29.35
Luminaire - 2x175W MBF - PARKWAY 2	\$0.00	\$154.34	\$154.34
Luminaire - 2x20W TF	\$0.00	\$12.57	\$12.57
Luminaire - 2x20W TF - WAVERLEY	\$0.00	\$12.57	\$12.57
Luminaire - 2x250W SON FLOODLIGHT	\$0.00	\$72.83	\$72.83
Luminaire - 2x26W TF MACQUARIE DEC. BALL	\$0.00	\$123.65	\$123.65
Luminaire - 2x400W MBF - PARKWAY 2	\$0.00	\$154.34	\$154.34
Luminaire - 2x400W MBI FLOODLIGHT	\$0.00	\$157.03	\$157.03
Luminaire - 2x400W SON FLOODLIGHT	\$0.00	\$171.46	\$171.46
Luminaire - 2x40W TF	\$0.00	\$29.69	\$29.69
Luminaire - 2x70W SON - BOURKE HILL	\$0.00	\$174.94	\$174.94
Luminaire - 2x80W MBF - BOURKE HILL	\$0.00	\$78.15	\$78.15
Luminaire - 3x400W MBF - PARKWAY 3	\$0.00	\$154.34	\$154.34
Luminaire - 400W MBF	\$0.00	\$32.54	\$32.54
Luminaire - 400W MBF - PARKWAY 1	\$0.00	\$72.83	\$72.83
Luminaire - 400W MBF FLOODLIGHT	\$0.00	\$82.73	\$82.73
Luminaire - 400W MBI - SMARTPOLE	\$0.00	\$2.07	\$2.07
Luminaire - 400W MBI FLOODLIGHT	\$0.00	\$55.84	\$55.84
Luminaire - 400W SON	\$0.00	\$32.60	\$32.60
Luminaire - 400W SON - PARKWAY 1	\$0.00	\$45.46	\$45.46
Luminaire - 400W SON FLOODLIGHT	\$0.00	\$66.52	\$66.52
Luminaire - 40W SOX	\$0.00	\$12.68	\$12.68

Asset	Prices to be levied in FY11		
	Maintenance	Capital	Total
Luminaire - 42W MBF SYLVANIA SUB ECO CFL	\$0.00	\$24.41	\$24.41
Luminaire - 4x1000W MBF	\$0.00	\$131.13	\$131.13
Luminaire - 4x20W TF	\$0.00	\$57.88	\$57.88
Luminaire - 4x20W TF - WAVERLEY	\$0.00	\$57.88	\$57.88
Luminaire - 4x250W SON	\$0.00	\$85.28	\$85.28
Luminaire - 4x40W TF	\$0.00	\$72.14	\$72.14
Luminaire - 4x40W TF - WAVERLEY	\$0.00	\$65.68	\$65.68
Luminaire - 4x600W SON	\$0.00	\$143.58	\$143.58
Luminaire - 500W MBI FLOODLIGHT	\$0.00	\$77.92	\$77.92
Luminaire - 50W MBF	\$0.00	\$12.73	\$12.73
Luminaire - 50W MBF - BOURKE HILL	\$0.00	\$78.15	\$78.15
Luminaire - 50W MBF - NOSTALGIA	\$0.00	\$76.53	\$76.53
Luminaire - 50W MBF - PLAIN	\$0.00	\$12.73	\$12.73
Luminaire - 50W MBF BOLLARD	\$0.00	\$42.42	\$42.42
Luminaire - 50W SON	\$0.00	\$12.33	\$12.33
Luminaire - 50W SON - BOURKE HILL	\$0.00	\$89.69	\$89.69
Luminaire - 50W SON - NOSTALGIA	\$0.00	\$30.09	\$30.09
Luminaire - 60W SOX	\$0.00	\$12.68	\$12.68
Luminaire - 700W MBF	\$0.00	\$37.18	\$37.18
Luminaire - 70W MBI	\$0.00	\$21.62	\$21.62
Luminaire - 70W MBI - MACQUARIE DEC. BALL	\$0.00	\$139.76	\$139.76
Luminaire - 70W SON	\$0.00	\$12.46	\$12.46
Luminaire - 70W SON - BOURKE HILL	\$0.00	\$89.69	\$89.69
Luminaire - 70W SON - GEC BOSTON 2	\$0.00	\$108.19	\$108.19
Luminaire - 70W SON - NOSTALGIA	\$0.00	\$82.33	\$82.33
Luminaire - 70W SON - PARKVILLE	\$0.00	\$108.19	\$108.19
Luminaire - 70W SON - REGAL/FLINDERS ENC	\$0.00	\$160.59	\$160.59
Luminaire - 70W SON BOLLARD	\$0.00	\$58.27	\$58.27
Luminaire - 70W SON FLOODLIGHT	\$0.00	\$24.00	\$24.00
Luminaire - 70W SON -PLAIN	\$0.00	\$12.46	\$12.46
Luminaire - 750W MBI FLOODLIGHT	\$0.00	\$77.92	\$77.92
Luminaire - 80W MBF	\$0.00	\$11.88	\$11.88
Luminaire - 80W MBF - BEGA+CURVE BRACKET	\$0.00	\$139.93	\$139.93
Luminaire - 80W MBF - BOURKE HILL	\$0.00	\$55.30	\$55.30
Luminaire - 80W MBF - GEC BOSTON 2	\$0.00	\$108.19	\$108.19
Luminaire - 80W MBF - NOSTALGIA	\$0.00	\$76.53	\$76.53
Luminaire - 80W MBF - PLAIN	\$0.00	\$11.88	\$11.88
Luminaire - 80W MBF - REGAL/FLINDERS ENC	\$0.00	\$154.22	\$154.22
Luminaire - 80W MBF - SYLVANIA SUBURBAN	\$0.00	\$12.07	\$12.07
Luminaire - 80W MBF BOLLARD	\$0.00	\$42.42	\$42.42
Luminaire - 80W MBF TOORAK	\$0.00	\$68.14	\$68.14
Luminaire - 90W SOX	\$0.00	\$58.68	\$58.68
Luminaire - INCANDESCENT	\$0.00	\$5.63	\$5.63
Luminaire - PRIVATE	\$0.00	\$1.38	\$1.38
Luminaire - TH FLOODLIGHT	\$0.00	\$151.79	\$151.79
Support - 2ND LIGHT NON-TRL	\$0.00	\$0.00	\$0.00
Support - 2ND LIGHT TRL	\$0.00	\$0.00	\$0.00
Support - BOLLARD	\$0.00	\$134.93	\$134.93
Support - COLUMN 10.5M-13.5M	\$0.00	\$263.20	\$263.20
Support - COLUMN 14M-15M	\$0.00	\$241.93	\$241.93
Support - COLUMN 2.5M-3.5M	\$0.00	\$211.91	\$211.91
Support - COLUMN 4-6.5M ORION WATER PIPE	\$0.00	\$229.96	\$229.96
Support - COLUMN 4M-6.5M	\$0.00	\$259.08	\$259.08
Support - COLUMN 7M-10M	\$0.00	\$251.91	\$251.91
Support - DECORATIVE COLUMN	\$0.00	\$277.82	\$277.82
Support - DEDICATED SUPPORT & CONDUCTOR	\$0.00	\$224.77	\$224.77
Support - HYDE PARK STANDARD	\$0.00	\$345.76	\$345.76
Support - MACQUARIE STANDARD	\$0.00	\$58.46	\$58.46
Support - MAST 15.5M-30M	\$0.00	\$253.89	\$253.89
Support - MAST 23M	\$0.00	\$253.89	\$253.89
Support - MAST 25M	\$0.00	\$253.89	\$253.89
Support - ORION DOUBLE ARM	\$0.00	\$39.70	\$39.70
Support - POLO 10.5M DECORATIVE 2M OUTREACH	\$0.00	\$79.00	\$79.00
Support - POLO 4.5M DECORATIVE 1.2 OUTREACH	\$0.00	\$79.00	\$79.00
Support - PRIVATE	\$0.00	\$0.00	\$0.00
Support - ROCKS STANDARD	\$0.00	\$201.80	\$201.80
Support - SMARTPOLE A	\$0.00	\$0.00	\$0.00
Support - SMARTPOLE AB	\$0.00	\$0.00	\$0.00
Support - SMARTPOLE B	\$0.00	\$0.00	\$0.00

Asset	Prices to be levied in FY11		
	Maintenance	Capital	Total
Support - SMARTPOLE C	\$0.00	\$0.00	\$0.00
Support - SMARTPOLE DOUBLE	\$0.00	\$10.52	\$10.52
Support - SMARTPOLE SINGLE LONG	\$0.00	\$10.52	\$10.52
Support - SMARTPOLE SINGLE SHORT	\$0.00	\$10.52	\$10.52
Support - SUSPENDED	\$0.00	\$0.00	\$0.00
Support - UNKNOWN	\$0.00	\$0.00	\$0.00
Support - WALL	\$0.00	\$0.00	\$0.00
Support - WOOD POLE NON-TRL	\$0.00	\$0.00	\$0.00
Support - WOOD POLE TRL	\$0.00	\$0.00	\$0.00

Attachment D: Completed Compliance Spreadsheet (Confidential)

Attachment E: Miscellaneous & Monopoly Charges and Emergency Recoverable Works

Tariff Classes

EnergyAustralia considers Miscellaneous Services, Monopoly Services, and Emergency Recoverable Works as each a separate tariff class. Table 1 sets out the customers which are allocated to each of these tariff classes.

Table 1: Tariff Classes and the customers that apply to them

Tariff Class	Applicable Customers
Miscellaneous Services	As defined in Appendix G1 of the Final Determination
Monopoly Services	As defined in Appendix G2 of the Final Determination
Emergency Recoverable Works	Where a person (the customer) is liable to another (which may include the DNSP) for damage which is the consequence of an act or omission of the person. This emergency work is undertaken by a DNSP to repair damage to the distribution system.

Table 2 and Table 4 (on next page) set out the services and charging parameters that apply to the monopoly and miscellaneous tariff classes. These charges are identical to Appendix H.3 and H.4 respectively of the AER's determination. Table 3 is the applicable labour rate to apply to monopoly services and is identical to that set out in Appendix H.4 of the AER's determination.

Table 2: Charges for Miscellaneous Services

Miscellaneous Service	\$
Special meter reading	\$44
Meter test	\$73
Supply of conveyancing information – desk inquiry	\$37
Supply of conveyancing information – field visit	\$73
Off peak conversion	\$59
Disconnection visit (acceptable payment received)	\$44
Disconnection at meter box	\$88
Disconnection at pole top / pillar Box	\$148
Rectification of illegal connection	\$221
Reconnection after business hours	\$95

Table 3: Applicable Labour Rates

Labour Class	Hourly rate
Admin R1	\$64
Design R2a	\$80
Inspector R2b	\$80
Engineer R3	\$96

Charges for emergency recoverable works

Under appendix H.5 of the AER's determination, the charge that a DNSP may levy for emergency recoverable works must not exceed the sum of the following:

- 1) 110 per cent of the costs (other than labour costs) actually incurred in providing the emergency recoverable works; and
- 2) the cost of labour actually used to undertake the emergency recoverable works determined by applying 150 per cent of the R2 labour rate for that labour (as referred to in Table 4 above.)

Pricing Principles (clause 6.18.5 of transitional chapter the National Electricity Rules)

Chapter 6 of the Transitional Rules require EnergyAustralia to demonstrate that each tariff class' expected revenue be between stand alone cost and avoidable of providing the service. The Rules also require EnergyAustralia to take into account the long run marginal cost for each charging parameter, and have regard for transaction costs and whether relevant tariff class is able to respond to price signals.

These Rule requirements, so far as they applicable to the monopoly, miscellaneous and emergency recovery tariff classes, should be considered as addressed through the implementation of the AER's determination. As noted above, the tariff Classes (and the services within them) and charging parameters are identical to those set out in Appendices G and H of the AER's Determination.

Table 4: Charges for Monopoly Services

Monopoly Service	Underground urban residential subdivision (vacant lots)				Rural overhead subdivisions and rural Extensions				Underground commercial and industrial or rural subdivisions (vacant lots - no development)				Commercial and Industrial Developments	Asset relocation or street lighting
Design Information	Up to 5 lots \$159 6 to 10 lots \$239 11 to 40 lots \$398 Over 40 lots \$478				R2 per hour				R2 per hour				R2 per hour	R2 or R3 per hour (See para 1.4.2 of Final Determination)
Design Certification	Up to 5 lots \$80 6 to 10 lots \$159 11 to 40 lots \$239 Over 40 lots \$318				1 - 5 poles \$80 6 -10 poles \$159 11 or more poles \$239				Up to 10 lots \$159 11 - 40 lots \$239 Over 40 lots \$478				R3 per hour	R2 or R3 per hour (See para 1.4.2 of Final Determination)
Design Rechecking	R2 per hour				R2 per hour				R2 per hour				R3 per hour	R2 or R3 per hour (See para 1.4.2 of Final Determination)
Inspection of service work (level 1 work)	Grade:	A	B	C	Grade:	A	B	C	Grade:	A	B	C	R2 or R3 per hour	R2 or R3 per hour (See para 1.4.2 of Final Determination)
	First 10 lots:	per lot	per lot	per lot	1-5 poles	per pole	per pole	per pole	First 10 lots	per lot	per lot	per lot		
	Next 40 lots:	\$40	\$96	\$200	6-10 poles	\$48	\$80	\$176	Next 40 lots	\$40	\$96	\$200		
	Remainder:	\$24	\$56	\$120	11+ poles (See para 1.4.2 of Final Determination)	\$32	\$56	\$120	Remainder	\$40	\$96	\$200		
Access Permit	Residential subdivisions: \$27 per lot combined fee				\$1181 maximum per access permit				\$1181 maximum per access permit				\$1181 maximum per access permit	\$1181 maximum per access permit
Substation Commissioning					\$886 per substation (See para 1.4.2 of Final Determination)				\$886 per substation (See para 1.4.2 of Final Determination)				\$886 per substation (See para 1.4.2 of Final Determination)	\$886 per substation (See para 1.4.2 of Final Determination)

Monopoly Service	Underground urban residential subdivision (vacant lots)	Rural overhead subdivisions and rural Extensions	Underground commercial and industrial or rural subdivisions (vacant lots - no development)	Commercial and Industrial Developments	Asset relocation or street lighting
Administration	Up to 5 lots \$193 6 - 10 lots \$258 11 - 40 lots \$322 Over 40 lots \$387	Up to 5 poles: \$193 6-10 poles: \$258 11 or more poles \$387	R1 per hour (max 6 hours)	R1 per hour (max 6 hours)	R1 per hour
Notice of Arrangement	\$193				
Re-inspection (level 1 and level 2 work)	R2 per hour (maximum 1 hour per level 2 reinspection)				
Re-inspection (service provider)	\$80 For the purpose of para 1.2(b), a DNSP may charge a fee that is less than this fee, but not a fee that is more than this fee.				
Access (Standby Person)	R1 per hour				
Authorisation	\$159				
Inspection of Service and Metering Work (Level 2 work)	All Service connections: A Grade : \$20 per NOSW B Grade: \$33 per NOSW C Grade: \$96 per NOSW (NOSW = Notification of Service Work)				
Site Establishment	\$139				

Attachment F: Notification of Climate Change Fund Contribution

Hon Frank Sartor AO MP

Minister for Climate Change and the Environment
Minister Assisting the Minister for Health (Cancer)



DOC10/4156

Mr George Maltabarow
Managing Director
EnergyAustralia
GPO Box 4009
SYDNEY NSW 2001



Dear Mr Maltabarow

I am writing in relation to EnergyAustralia's annual contribution to the Climate Change Fund.

I am currently consulting with the Treasurer and the Minister for Energy about the making of an order under section 34J of the *Energy and Utilities Administration Act 1987* (the Act) to require distribution network service providers to make a contribution to the Climate Change Fund for the financial year 2010/11.

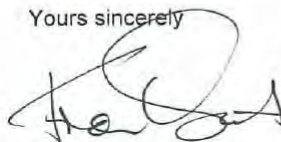
I understand that EnergyAustralia is required to submit its distribution network charges pricing proposal to the Australian Energy Regulator for the 2010/11 regulatory year by 30 April 2010. I am therefore advising you that I may make an order under the Act to provide that EnergyAustralia is to make a total annual contribution to the Fund for the financial year 2010/11 of \$70,996,779. The annual contribution will be payable in four quarterly instalments on 1 August 2010, 1 November 2010, 1 February 2011 and 1 May 2011.

As in previous years, I am requesting that EnergyAustralia recover through distribution network charges no more than 25 per cent of the contributions from residential customers.

I would appreciate any views you may have prior to the making of the order.

Thank you for your assistance.

Yours sincerely



25 MAR 2010

Frank Sartor MP

Level 35, Governor Macquarie Tower, 1 Farrer Place, SYDNEY NSW 2000
Telephone: +61 2 9228 5811 Facsimile: +61 2 9228 5499

Attachment G: Procedure for Assignment and Re-assignment of Tariffs

This document demonstrates EnergyAustralia’s approach for assigning a new network connection to a tariff and re-assigning an existing network connection from one tariff to another and where an assignment or re-assignment is between tariff classes, this will occur in accordance with Appendix A of the AER network determination.

Default Tariffs and Tariff Transfers

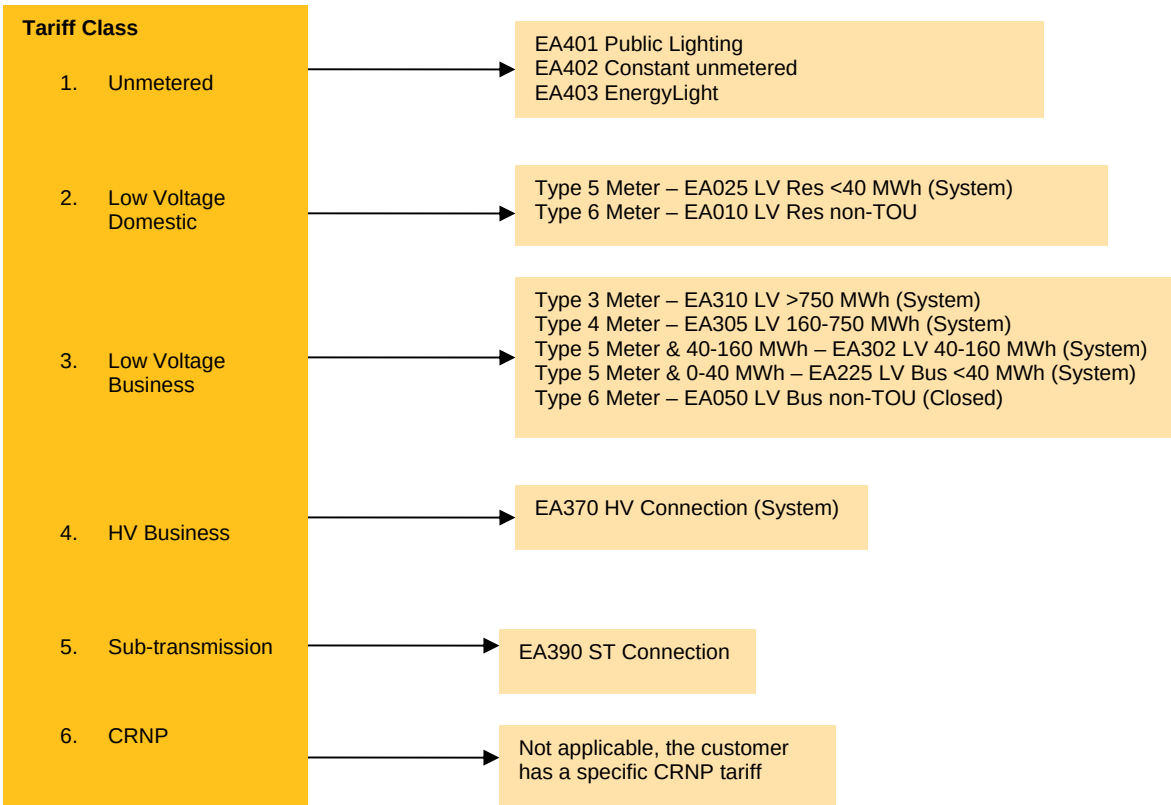
A default tariff is the network tariff to which EnergyAustralia assigns a customer based on its tariff assignment and re-assignment policies. A customer may move away from its assigned default tariff after 12 months of load history has been accumulated. The customer, or their retailer, can then apply to transfer to

an alternative network price using the Network Price Application Form available in the ES7 “Application of Network Use of System Charges” document. Applications for tariff transfers are usually the result of annual consumption changes, or requests for a substation tariff (discussed below). Backdating of new network prices is not permitted and customers may only have their prices changed once in a 12 month period.

Tariff Assignment and Re-assignment for Existing Customers

EnergyAustralia’s default tariff allocation procedure for existing customers is shown in Figure 1. It is noted that the procedure for assigning customers to tariff classes is contained in Section II of the main Pricing Proposal.

Figure 1 Decision Tree of Assigning Existing Customers to Default Tariffs



Cost Reflective Network Price (CRNP) customers are defined as customers with a demand in excess of 10MW on at least three half hour periods within a 12 month period. Site specific prices are calculated each year for these customers. If a customer falls below this threshold, then a default network is applied, consistent with the decision tree above.

It can be seen from figure 1 that the tariffs applicable to Type 6 metering installations are no longer available. It is EnergyAustralia’s policy that all new and upgraded connections (eg single phase to multiple phase connections) must install a Type 5 or better (Types 1 to 4) meter.

Tariff Assignments for New Customers

New connections (where there is no previous load history) are assigned to a default tariff which is based on tariff class (as defined in Part 2 of this Pricing Proposal), metering installation type and metering phase. Table 1 shows the way in which new customers are allocated to

default tariffs. Please note that this should be read in conjunction with EnergyAustralia's procedure for assigning customers to tariff classes, as explained in the main Pricing Proposal.

Table 1: The assignment of New Customers to Default Tariffs

Tariff Class		Metering Installation Type (if applicable)	Connection phase (if applicable)	Default Principal Network Price Code	Default Principal Network Price Name
Low Voltage	Residential			EA025	LV Res < 40 MWh (System)
LV Business	Non-Residential	3		EA310	LV >750 MWh (System)
		4		EA305	LV 160-750 MWh (System)
		5	Three phase	EA302	LV 40-160 MWh (System)
			Single phase	EA225*	LV Bus <40 MWh (System)*
HV Business				EA370	HV Connection (System)
Sub-transmission				EA390	ST Connection
Unmetered		7		EA401	Public Lighting
				EA402	Constant Unmetered

It is important to note that new residential connections (where there is no load history) will be assigned to the default tariff (EA25), but may at their discretion elect to be re-assigned to the inclining block tariff (EA10) if they satisfy the criteria for EnergyAustralia's opt-out policy. Similarly, existing residential connections on the default tariff (EA25) can apply to be re-assigned to the inclining block tariff (EA10).

The assessment of the customer's usage is based on the most recent 12 months of history. However if a customer's consumption has fallen because of vacancy of one month or more during the previous 12 months, the customer is excluded from a potential tariff re-assignment. This is to avoid making tariff transfers based on load data that is not fairly represent a customer's annual usage.

System of Assessment and Review

A review of a customer's network tariff is carried out annually to assess if the tariff is still correct given potential changes in annual usage and meter type. A change in voltage is treated as a new connection. EnergyAustralia's network tariff policy is generally aligned to consumption bands. For example, network tariff EA225 LV <40 MWh (System) applies to 0 to 40 MWh pa business customers. Above this tariff EA302 LV 40-160 MWh (System) applies to customers using 40 to 160 MWh pa.

EnergyAustralia applies a tolerance of ± 20 percent around tariff thresholds. If a customer on EA225 LV <40 MWh (System) were to consume 49 MWh pa, they would be reassigned immediately to the new tariff, being more than 20 percent above the 40 MWh threshold. However, if they are using 43 MWh pa, the customer falls within the band tolerance of 40 MWh ± 8 MWh so their tariff re-assignment is deferred. If the same customer is then found to still consume more than 40 MWh after two years, the threshold bands do not apply, and the customer will be reassigned from network tariff EA225 to EA302. The relevant retailer is notified of the impending change, and the customer is re-assigned to the new network tariff.

Substation Tariffs

EnergyAustralia has decided to make redundant the substation network tariffs applying to low voltage supplies from 1 July 2010. The substation tariff applying high voltage supplies will continue to be offered in FY11.

A substation price will be applied to high voltage supplies, which satisfy the following criteria:

- A single customer's mains are directly connected to the EnergyAustralia zone substation HV busbar;
- The customer owns, operates and maintains these mains;
- The customer can demonstrate that they have capitally contributed to the construction of the substation.

If approved, the price change will apply from the start of the next billing period (typically monthly) following the date of receipt of the price change application.

Standby rates

The LV and HV standby tariffs (with codes EA325 and EA350 respectively) are applied to customers who have a stand-by connection at their premises. The two standby tariffs consist of a large fixed charge and standard energy rates. The fixed charge is designed to recover the

cost of the feeder, even while no capacity is being used. Whether or not the customer uses the supply, the customer will be charged a fixed daily charge for the provision of the supply. This will simplify billing for these supplies. It will also act as a clear signal of the cost to EnergyAustralia of providing the standby supply capacity. This is consistent with our strategy to keep tariffs simple and cost reflective where possible, and to send appropriate pricing signals to our customers.

No DUOS charges for the export of energy

Where a customer with distributed generation facilities is able to supply their own load and also generate energy into the network, no charges are applied to the energy exported. If the customer is purely a generating source (with no local load) then no network tariff applies. Network tariffs are only applied where load is present.

Public lighting

EnergyAustralia's public lighting network use of system tariff (EA401) consists of an energy rate, but does not have a fixed charge. This tariff is available for metered and unmetered connections for a public lighting service.

**Attachment H: Statement from NERA Economic Consulting
(Confidential)**

**Attachment I: Statement from PriceWaterhouseCoopers Australia
(Confidential)**

Attachment J: Confirmation of TransGrid's Transmission Charges for FY11



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Jane Smith
Executive Manager - Pricing and Regulation
EnergyAustralia
GPO Box 4009 Sydney NSW 2001 Australia

Dear Jane,

Re: NSW Transmission Prices for 2010-11

The purpose of this letter is to provide formal confirmation of the NSW transmission use of system prices (TUOS) to apply to EnergyAustralia for 2010-11.

The 2010-11 TUOS price reflects the recovery of regulated revenues for all NSW transmission network service providers (TNSPs), being TransGrid, EnergyAustralia, Country Energy and Directlink Joint Venture and have been calculated by TransGrid as the co-ordinating TNSP for the NSW Market region.

This letter presents the following information:

- Attachment 1 – The NSW transmission use of system prices to apply to EnergyAustralia for 2010-11;
- Attachment 2 – Contains a table of financial transfers between the TNSPs for 2010-11 developed by TransGrid as the co-ordinating TNSP for the NSW market region.
- Attachment 3 – Explains the major drivers for the 2010-11 TUOS price.

It is acknowledged that the 2010-11 TUOS prices for load customers and distributors have increased overall by more than expected Consumer Price Index level movements. This can largely be attributed to:

- The emergence of negative NSW intra-regional settlement residues in Quarter 2 and 3 of 2009-10, resulting in an approximate \$36M under-collection for 2009-10;
- It is expected that the current trend of negative NSW intra-regional settlement residues would continue through to 2010-11 resulting in a reduction by approximately \$35M in the NSW intra-regional settlement residues forecast for 2010-11;
- The Australian Competition Tribunal ("The Tribunal") variation of the Australian Energy Regulator's (AER) TransGrid transmission determination 2009-10 to 2013-14; and
- The Tribunal's variation of the AER's New South Wales distribution determination 2009-10 to 2013-14 as it applied to EnergyAustralia's Transmission Regulatory Asset Base.

The revenue generated from the NSW Transmission Prices for 2010-11 will, in part, be allocated to the aggregate annual revenue requirement to support TransGrid's capital expenditure program over the next five years of approximately \$2.6 billion. This is required to maintain TransGrid's transmission network reliability at more than 99.999% in line with good practice requirements for a transmission system supporting a modern growing economy.

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This program includes network augmentation driven by reliability obligations expected from growth in the NSW economy, in general, and growth in Sydney's population of 1.1 million over the next 25 years, in particular. The program also provides for extensive network replacement of TransGrid's mature asset base which was established in the 1950s and 1960s and is now reaching the end of its useful life.

In anticipation of the close focus the 2010-11 TUOS prices could receive from affected parties, TransGrid has prepared Attachment 3 to inform internal and external stakeholders of the major drivers for the 2010-11 TUOS price. For this reason TransGrid would be happy for EnergyAustralia to distribute Attachment 3 to any stakeholders who EnergyAustralia feel would be appropriate.

TransGrid would also be happy to meet with any EnergyAustralia internal or external stakeholders who wish to discuss aspects of TransGrid's transmission pricing. This could be arranged by contacting myself on 02 9284 3434 (email: philip.gall@transgrid.com.au), or Sean Buggy on 02 9284 3072 (email: sean.buggy@transgrid.com.au).

Yours sincerely

Phil Gall 12/4/10

Philip Gall
Manager/Regulated Transmission Access

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22. Published Price List



TransGrid's Transmission Prices
1 July 2010 to 30 June 2011

(All rates quoted are inclusive of Australian Goods and Services Tax (GST))

Energy Australia

Customer Charges

Common service and general charges

These rates apply at all connection points

	Energy Rate (c/kWh)	Contract Rate (\$/kW/month)
Common Service Charge	0.3122	1.1041
General Charge	0.6451	2.2812

Usage and exit charges

Customer	Connection Point	Exit (\$/day)	Fixed (\$/day)	Peak (c/kWh)	Usage Charge			Demand (\$/kW/mth)
					Shoulder (c/kWh)	Off-peak (c/kWh)		
Energy Australia	957 Tee	0.00	0.00	0.0000	0.0000	0.0000		1.1995
Energy Australia	Beaconsfield W 132	560.02	0.00	0.0000	0.0000	0.0000		2.5483
Energy Australia	Beresfield 132kV	3841.47	0.00	0.0000	0.0000	0.0000		1.3510
Energy Australia	Bunnerong 33	17243.62	0.00	0.0000	0.0000	0.0000		4.4244
Energy Australia	Canterbury 33	9665.30	0.00	0.0000	0.0000	0.0000		3.0477
Energy Australia	Charm Haven 11	5351.33	0.00	0.0000	0.0000	0.0000		1.2626
Energy Australia	Chullora 132	0.00	0.00	0.0000	0.0000	0.0000		3.5913
Energy Australia	Drummoyne 11	6293.47	0.00	0.0000	0.0000	0.0000		8.0301
Energy Australia	Gosford 33kV	1276.69	0.00	0.0000	0.0000	0.0000		2.1018
Energy Australia	Gosford 66kV	8307.58	0.00	0.0000	0.0000	0.0000		2.0375
Energy Australia	Green Square 11kV	6344.68	0.00	0.0000	0.0000	0.0000		3.0636
Energy Australia	Haymarket 132	882.49	0.00	0.0000	0.0000	0.0000		3.0838
Energy Australia	Homebush Bay 11	6897.67	0.00	0.0000	0.0000	0.0000		2.8310
Energy Australia	Kurri 11	2982.35	0.00	0.0000	0.0000	0.0000		1.4811
Energy Australia	Kurri 132	0.00	0.00	0.0000	0.0000	0.0000		1.3876
Energy Australia	Kurri 66	1791.83	0.00	0.0000	0.0000	0.0000		3.0849
Energy Australia	Lane Cove 132	0.00	0.00	0.0000	0.0000	0.0000		3.5058
Energy Australia	Liddell 330	0.00	0.00	0.0000	0.0000	0.0000		0.4798
Energy Australia	Macquarie Park 11	6344.68	0.00	0.0000	0.0000	0.0000		5.9359
Energy Australia	Marrickville 11	5632.61	0.00	0.0000	0.0000	0.0000		2.9186
Energy Australia	Mason Park 132	0.00	0.00	0.0000	0.0000	0.0000		2.8569
Energy Australia	Meadowbank 11	9025.46	0.00	0.0000	0.0000	0.0000		3.8970
Energy Australia	Munmorah 132	0.00	0.00	0.0000	0.0000	0.0000		1.2009
Energy Australia	Munmorah 330	0.00	0.00	0.0000	0.0000	0.0000		0.8697
Energy Australia	Muswellbrook 132	3475.13	0.00	0.0000	0.0000	0.0000		0.5621
Energy Australia	Newcastle 132	738.11	0.00	0.0000	0.0000	0.0000		0.8814
Energy Australia	Ourimbah 33	4237.46	0.00	0.0000	0.0000	0.0000		1.6598
Energy Australia	Peakhurst 33	9471.59	0.00	0.0000	0.0000	0.0000		2.0842
Energy Australia	Pymont 132	2301.18	0.00	0.0000	0.0000	0.0000		3.8649
Energy Australia	Pymont 33	10820.11	0.00	0.0000	0.0000	0.0000		5.0431
Energy Australia	Rozelle 33kV	1387.44	0.00	0.0000	0.0000	0.0000		5.7012
Energy Australia	Somersby 132	5294.67	0.00	0.0000	0.0000	0.0000		1.8221
Energy Australia	St Peters 11	6841.26	0.00	0.0000	0.0000	0.0000		2.7858
Energy Australia	Sydney East 132	1968.44	0.00	0.0000	0.0000	0.0000		1.9717
Energy Australia	Sydney North 132	1331.57	0.00	0.0000	0.0000	0.0000		2.4145
Energy Australia	Sydney South 132	802.82	0.00	0.0000	0.0000	0.0000		1.7920
Energy Australia	Tomago 33	6681.75	0.00	0.0000	0.0000	0.0000		1.5113
Energy Australia	Tuggerah 132	250.25	0.00	0.0000	0.0000	0.0000		2.2080
Energy Australia	VAW 132	0.00	0.00	0.2596	0.2596	0.2596		0.0000
Energy Australia	Waratah West 132	872.70	0.00	0.0000	0.0000	0.0000		1.2776
Energy Australia	Warrabrook 132	0.00	0.00	0.0000	0.0000	0.0000		1.2776
Energy Australia	West Gosford 132	4910.98	0.00	0.0000	0.0000	0.0000		1.7280
Energy Australia	Wyong 11	5351.33	0.00	0.0000	0.0000	0.0000		1.4912

FINANCIAL TRANSFERS BETWEEN TRANSMISSION NETWORK SERVICE PROVIDERS IN NSW

1 July 2010 to 30 June 2011

GST = 10%

Transfer from	Transfer to	Annual Payment (Excluding GST)	Monthly Payment	
			(Excluding GST)	(Including GST)
Country Energy	Directlink JV	\$2,909,939.26	\$242,494.94	\$266,744.43
Country Energy	EnergyAustralia	\$194,943.42	\$16,245.29	\$17,869.81
Country Energy	TransGrid	\$14,349,949.62	\$1,195,829.14	\$1,315,412.05
	Subtotal:	\$17,454,832.30	\$1,454,569.36	\$1,600,026.29
EnergyAustralia	Country Energy	\$0.00	\$0.00	\$0.00
EnergyAustralia	Directlink JV	\$1,149,969.70	\$95,830.81	\$105,413.89
EnergyAustralia	TransGrid	\$133,876,673.44	\$11,156,389.45	\$12,272,028.40
	Subtotal:	\$135,026,643.15	\$11,252,220.26	\$12,377,442.29
TransGrid	Country Energy	\$0.00	\$0.00	\$0.00
TransGrid	Directlink JV	\$8,972,721.04	\$747,726.75	\$822,499.43
TransGrid	EnergyAustralia	\$45,209,522.48	\$3,767,460.21	\$4,144,206.23
	Subtotal:	\$54,182,243.52	\$4,515,186.96	\$4,966,705.66
Net Transfer - EA to TG		\$88,667,150.96	\$7,388,929.25	\$8,127,822.17

ATTACHMENT 3



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SUBJECT: REVENUE AND TRANSMISSION PRICES FOR 2010/11

Purpose

This explanation of the 2010-11 TUOS revenue and price drivers serves as an Attachment to TransGrid's notification of the 2010-11 NSW Transmission Prices. The 2010-11 TUOS price reflects the recovery of regulated revenues for all NSW transmission network service providers being TransGrid, EnergyAustralia and the Directlink Joint Venture and have been calculated by TransGrid as the co-ordinating TNSP for the NSW Market region. In so doing TransGrid has complied with the requirements of the Transmission Pricing Methodology approved by the Australian Energy Regulator in April 2009. The purpose of this explanation is to inform internal and external stakeholders of NSW load customers and distributors.

Conditions of use

Please note this explanation has been prepared to address the overall 2010-11 TUOS revenue and price for NSW load customers and distributors, and is not designed to explain 2010-11 TUOS price movements for individual NSW load customers and distributors. Such a detailed explanation can be provided on request. Any portion of the 2010-11 TUOS price passed on by distributors to distribution customers is done at the discretion of the distributor with no input from TransGrid.

A majority of the 2009-10 and 2010-11 data refers to forecast data. Because the actual revenue collected is dependent on actual energy consumption and demand, there could be a departure between the 2010-11 TUOS forecast revenue paid by NSW load customers and distributors and the actual TUOS revenue paid.

Introduction

TransGrid is the owner, operator and manager of the major part of the NSW's high-voltage transmission network connecting generators, distributors and major end users in NSW and the ACT. The balance of the network is owned, operated and managed by EnergyAustralia, Country Energy and the Directlink Joint Venture in their local areas.

It is acknowledged that the 2010-11 TUOS prices for load customers and distributors have increased overall by more than expected Consumer Price Index level movements. This can largely be attributed to:

- The emergence of negative NSW intra-regional settlement residues in Quarter 2 and 3 of 2009-10, resulting in an approximate \$36M under-collection for 2009-10;
- It is expected that the current trend of negative NSW intra-regional settlement residues would continue through to 2010-11 resulting in a reduction by approximately \$35M in the NSW intra-regional settlement residues forecast for 2010-11;
- The Australian Competition Tribunal ("The Tribunal") variation of the Australian Energy Regulator's (AER) TransGrid transmission determination 2009-10 to 2013-14; and

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- The Tribunal's variation of the AER's New South Wales distribution determination 2009–10 to 2013–14 as it applied to EnergyAustralia's Transmission Regulatory Asset Base.

The revenue generated from the NSW Transmission Prices for 2010–11 will, in part, be allocated to the aggregate annual revenue requirement to support TransGrid's capital expenditure program over the next five years of approximately \$2.6 billion. This is required to maintain TransGrid's transmission network reliability at more than 99.999% in line with good practice requirements for a transmission system supporting a modern growing economy.

This program includes network augmentation driven by reliability obligations expected from growth in the NSW economy, in general, and growth in Sydney's population of 1.1 million over the next 25 years, in particular. The program also provides for extensive network replacement of TransGrid's mature asset base which was established in the 1950s and 1960s and is now reaching the end of its useful life.

Summary explanation of the major drivers of the overall 2010–11 TUOS revenue and price

As can be seen in Table 1 the 2010–2011 forecast revenue required for all NSW TNSPs is \$904M.

2009–2010 Forecast Revenue \$M	2010–2011 Forecast Revenue \$M	Movement \$M
750	904	154

Table 1

The increase of \$154M is explained in Table 2 below as being firstly a decrease in the offsets available to reduce TUOS combined with secondly an increase in the TUOS requirement. See Table 3 for detail.

Reduction in TUOS Offset Movement (\$M)	50
TUOS Requirement Movement (\$M)	104
Movement (\$M)	154

Table 2

Figure 1 below demonstrates that offsets available to reduce TUOS revenue and pricing for customer loads and distributors is trending to zero when in the years leading up to 2009–10 the available offset was around the \$150M mark.

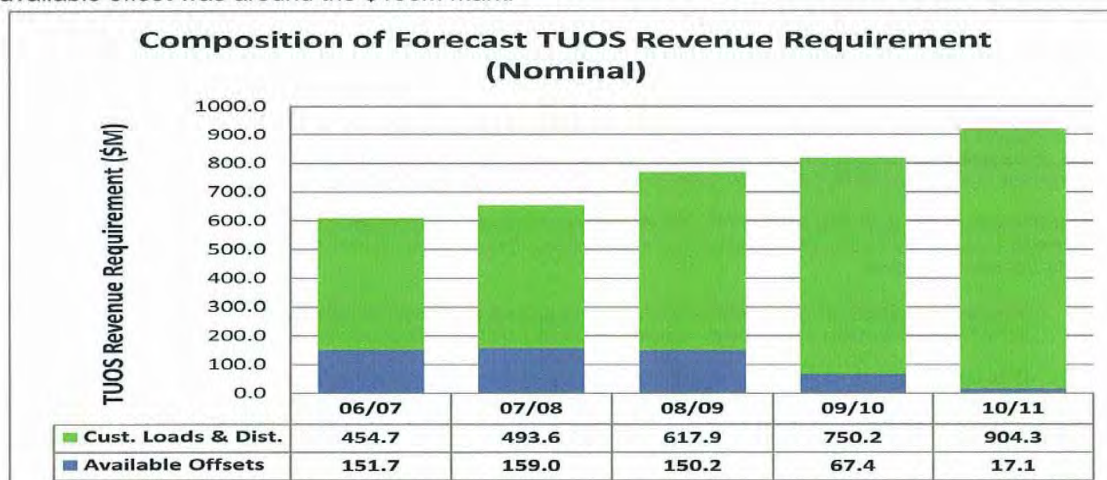


Figure 1

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Detailed explanation of the overall 2010-11 TUOS revenue and price drivers

Items available to offset TUOS Requirement	2009-2010 Forecast Offset \$M	2010-2011 Forecast Offset \$M	Movement \$M	Note
AEMO – Intra-Reg Set Residues	27.0	-8.0	-35.0	1
AEMO – SRA Proceeds	53.0	56.8	3.8	
Under Collection	-25.9	-37.5	-11.6	2
Generators	13.3	5.8	-7.5	3
Sub Total	67.4	17.1	-50.3	

TUOS Requirement Items ¹	2009-2010 TUOS Requirement \$M	2010-2011 TUOS Requirement \$M	Movement \$M	Note
TransGrid Transmission Revenue	665.4	735.2	69.8	4
EnergyAustralia Trans. Revenue	139.6	173.2	33.6	5
DirectLink Transmission Revenue	12.6	13.0	0.4	
Sub Total	817.6	921.5	103.8	
NSW load customers & distributors	750.2	904.3	154.1	

Table 3

Note Explanation

1. Intra-regional residues settled by TransGrid with AEMO

The Intra-Regional Settlement Residue is formed by subtracting the dollar amount of the NSW energy generated and imported to NSW, from NSW purchased energy within the National Electricity Market (NEM) operated by AEMO. The differences that occur are related to the Marginal Loss Factors set for both NSW generator and load connection points by AEMO and by the dynamic losses determined as part of the AEMO dispatch process for the import to NSW. More information can be sourced from the AEMO website.

In accordance with the National Electricity Rules (NER), AEMO allocates Intra-Regional Residues to the TNSP for NEM financial settlement. In turn, under the AER approved Transmission Pricing Methodology, the distribution of the Intra-Regional Residues will be used by TransGrid to offset the TUOS requirement.

Prior to 2009-10 the Intra-Regional Settlement Residue represented a significant credit that could be offset against the TUOS requirement, reducing the TUOS bills of NSW load customers and distributors. In fact, the 2009-10 Intra-regional residue was forecast at \$27M credit. However, commencing in the second and third quarter of this financial year, there was a step change in the size and frequency of NSW negative intraregional residues.

¹ This is the annual revenue cap following AER approved adjustments such as performance incentive payments, pass throughs (or refunds of) and other items.

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For financial year to mid March 2010, TransGrid's net settlement of intraregional residues totaled around \$6M in debit. In 2009 for the same time the intraregional residue settlement totaled \$17.79M in credit. Much of the \$6M debit balance can be attributed to eight settlement weeks ranging from \$600K to \$4.4M in debit, yet in the four previous financial years there was only one settlement week with a material negative intraregional residue.

TransGrid has written to, and met with AEMO, to seek AEMO's assistance in understanding this step change in NSW intra-regional residues. AEMO's preliminary assessment is that the NSW intra-regional residues are being calculated in accordance with the NER but AEMO has identified potential issues for investigation, and intends to publish the results of this investigation in the coming months.

2. Under Collection

This item represents the reconciliation between the 2009-10 TransGrid Revenue Cap against the 2009-10 expected TransGrid collection.

	\$M
Expected Collection 2009-10	629.2
TG 2009-10 Revenue Cap	-665.4
Under Collection 2009-10^	-36.2
Interest -Under Collection 2009-10	-1.3
Total Under Collection 2009-10	-37.5

^The major component comprises the emergence, in 2009-10, of negative intraregional residues in Quarter 2 and Quarter 3 mentioned in Note 1. The \$36M under collection occurred largely because the 2009-10 intraregional residues was forecast as \$27M but actual collection was approximately \$8M deficit (\$27M expected to be collected + \$8M paid out = \$35M under collection from intra-regional residues)

3. Generators

This item represents the TUOS charge paid by Generators. NER clause 11.6.11 becomes effective as at 2010-11 and contains revised arrangements for transitioning from prescribed transmission connection services to negotiated transmission connection services, resulting in a reduction of the TUOS charge levied on Generators.

4. TransGrid Revenue Cap

On 25 November 2009 the Australian Competition Tribunal ("The Tribunal") made orders varying the AER's TransGrid transmission determination 2009-10 to 2013-14

5. EnergyAustralia Revenue Cap

On 25 November 2009 the Tribunal made orders varying the AER's New South Wales distribution determination 2009-10 to 2013-14 as it applied to EnergyAustralia's Transmission Regulatory Asset Base

Any questions arising from this attachment may be directed to Sean Buggy, Customer Account Executive on 02 9284 3072 or email sean.buggy@transgrid.com.au.

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Attachment K: Economic Cost Model (Confidential)

See attached file

Attachment L: Proposed TUOS Transitional Arrangements (Confidential)